VORON



VORON STEALTHBURNER

We build space shuttles with gardening tools
so anyone can have a space shuttle of their own.

VERSION 2023-07-07



Before you begin on your journey, a word of caution.

In the comfort of your own home you are about to assemble a robot. This machine can maim, burn, and electrocute you if you are not careful. Please do not become the first VORON fatality. There is no special Reddit flair for that.

Please, read the entire manual before you start assembly. As you begin wrenching, please check our Discord channels for any tips and questions that may halt your progress.

Most of all, good luck!

THE VORON TEAM

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PART PRINTING GUIDELINES

The Voron Team has provided the following print guidelines for you to follow in order to have the best chance at success with your parts. There are often questions about substituting materials or changing printing standards, but we recommend you follow these:

3D PRINTING PROCESS

Fused Deposition Modeling (FDM)

MATERIAL

ABS

LAYER HEIGHT

Recommended: 0.2mm

EXTRUSION WIDTH

Recommended: Forced 0.4mm

INFILL TYPE

Grid, Gyroid, Honeycomb, Triangle or Cubic

INFILL PERCENTAGE

Recommended: 40%

WALL COUNT

Recommended: 4

SOLID TOP/BOTTOM LAYERS

Recommended: 5

PRINT IT FORWARD (PIF)

Often times community members that have issues printing ABS will bootstrap themselves into a VORON using our Print It Forward program. This is a service where approved members with VORON printers can make you a functional set of parts to get your own machine up and running. Check Discord if you have any interest in having someone help you out.

FILE NAMING

By this time you should have already downloaded our STL files from the Voron GitHub. You might have noticed that we have used a unique naming convention for the files. This is how to use them.

PRIMARY COLOUR

Example z_joint_lower_x4.stl

These files will have nothing at the start of the filename.

ACCENT COLOUR

Example [a]_tensioner_left.stl

We have added "[a]" to the front of any STL file that is intended to be printed with accent color. The parts are marked with a heart in the manual.

CLEAR/TRANSPARENT

Example [c]_led_diffuser.stl

We have added "[c]" to the front of any STL file that is intended to be printed with clear or transparent filament (cloudy is better than clear; white or light grey may also work in a pinch).

OPAQUE

Example [o]_led_diffuser_mask.stl

We have added "[o]" to the front of any STL file that is intended to be printed with an opaque filament (blocks light, dark colours and blacks).

QUANTITY REQUIRED

Example [a]_z_belt_clip_lower_x4.stl

If any file ends with "_x#", that is telling you the quantity of that part required to build the machine.

HOW TO GET HELP

If you need assistance with your build, we're here to help. Head on over to our Discord group and post your questions. This is our primary medium to help VORON Users and we have a great community that can help you out if you get stuck.



https://discord.gg/voron

REPORTING ISSUES

Should you find an issue in the documentation or have a suggestion for an improvement please consider opening an issue on GitHub (https://github.com/VoronDesign/Voron-Stealthburner/issues). When raising an issue please include the relevant page numbers and a short description; annotated screenshots are also very welcome. We periodically update the manual based on the feedback we get.

THIS IS JUST A REFERENCE

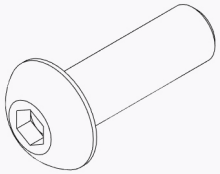
This manual is designed to be a simple reference manual. Building a Voron can be a complex endeavour and for that reason we recommend downloading the CAD files off our Github repository if there are sections you need clarification on. It can sometimes be easier to follow along when you have the whole assembly in front of you.



https://github.com/vorondesign



https://docs.vorondesign.com



BUTTON HEAD CAP SCREW (BHCS)

Metric fastener with a domed shape head and hex drive. Most commonly found in locations where M5 fasteners are used.

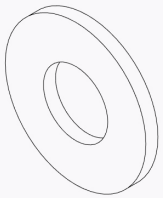
ISO 7380-1



FLAT HEAD CAP SCREW (FHCS)

Metric fastener with a flat head and hex drive.

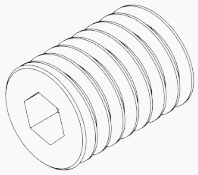
ISO 10642



WASHER

Usually stamped from sheet metal this type of spacer is not as consistent in thickness as the shims are. Only used in M3 size.

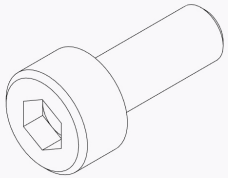
DIN 125



SET SCREW

Small headless screw with an internal drive. Used in pulleys and other gears. Also called a grub screw.

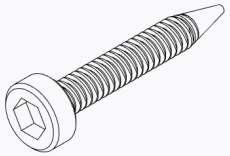
ISO 4026



SOCKET HEAD CAP SCREW (SHCS)

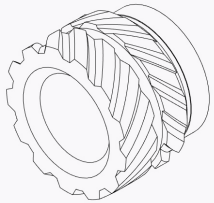
Metric fastener with a cylindrical head and hex drive. The most common fastener used on the Voron.

ISO 4762



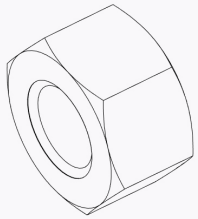
SELF TAPPING SCREW

Fastener with a pronounced thread profile that is screwed directly into plastic.



HEAT SET INSERT

Heat inserts with a soldering tip so that they melt the plastic when installed. As the plastic cools, it solidifies around the knurls and ridges on the insert for excellent resistance to both torque and pull-out.

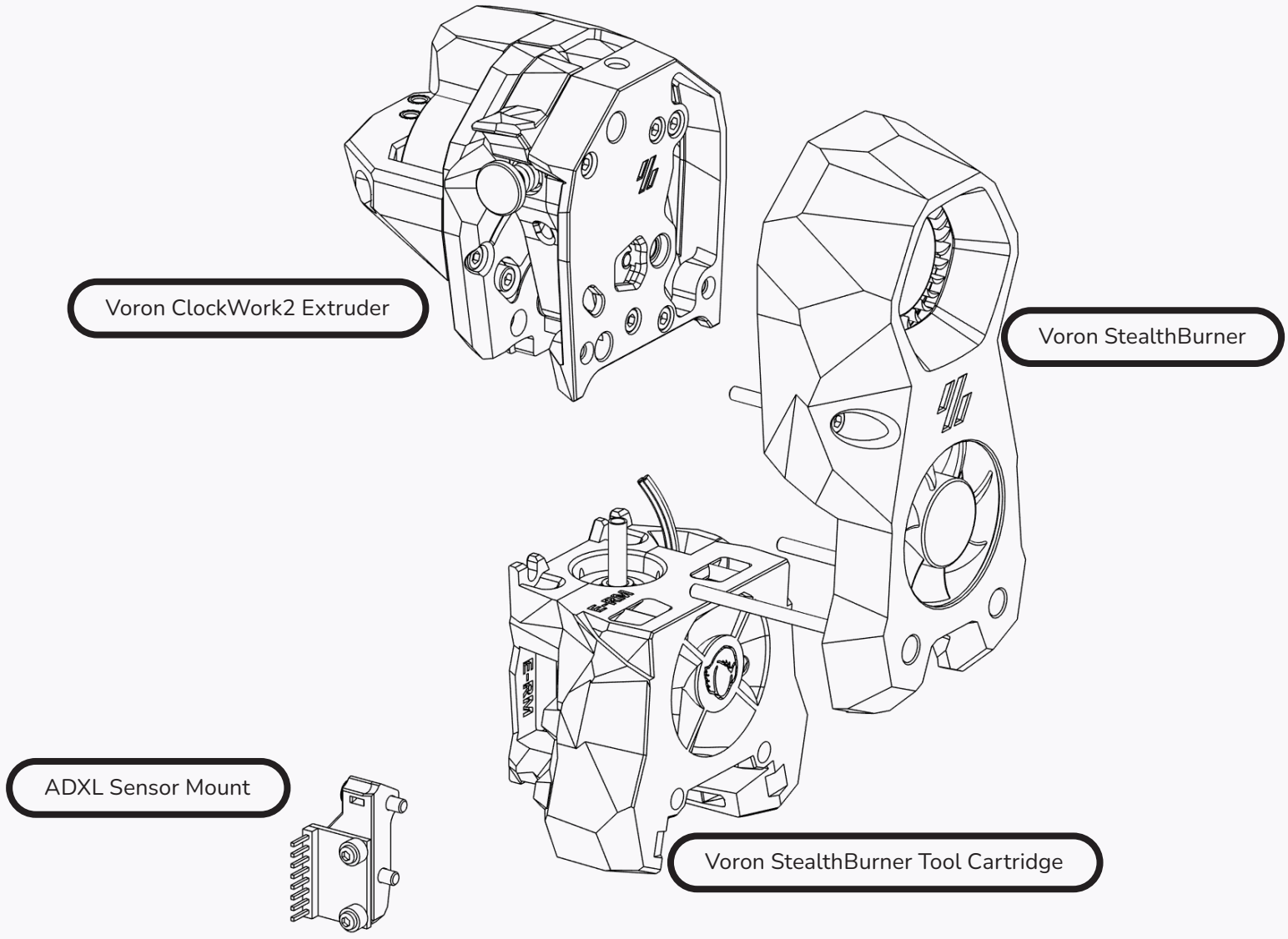


HEX NUT

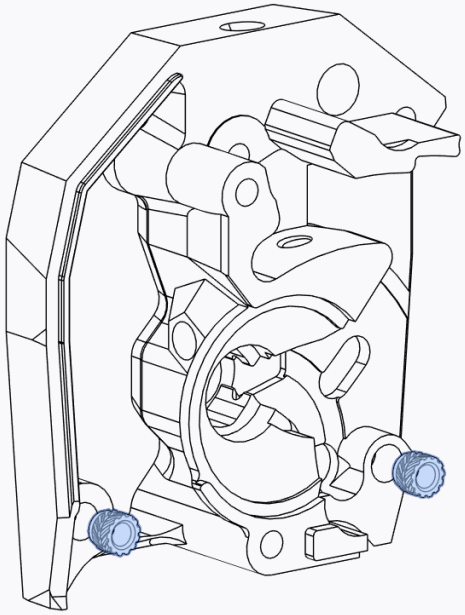
Hex nuts couple with bolts to create a tight, secure joint.

ISO 4032




Voron ClockWork2 Extruder
Voron StealthBurner
ADXL Sensor Mount
Voron StealthBurner Tool Cartridge



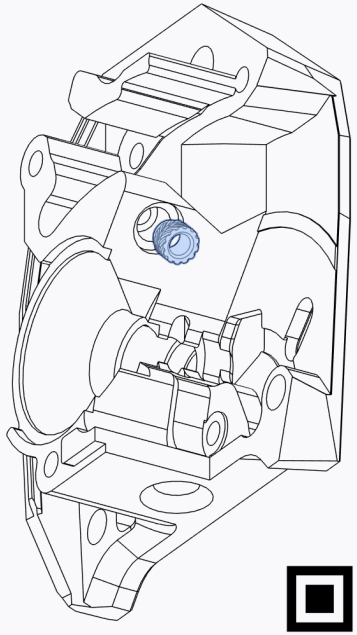


HEAT SET INSERTS

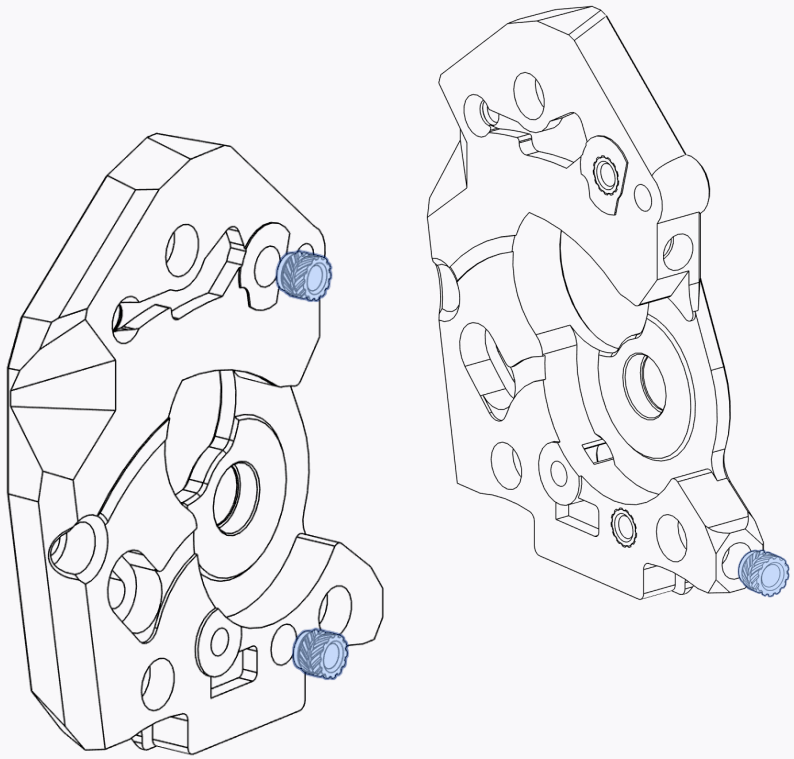
This design relies heavily on heat set inserts. Make sure you have the proper inserts (check the hardware reference for a close-up picture and the Sourcing Guide for dimensions).

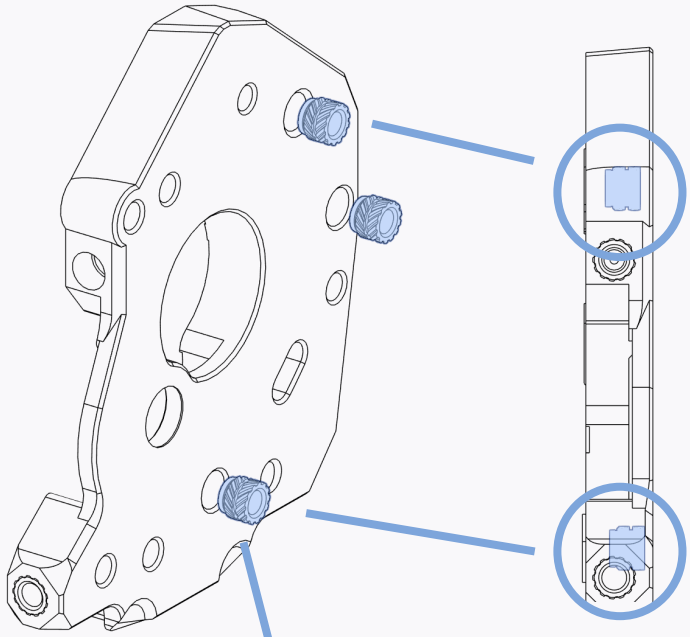
If you've never worked with heat set inserts before we recommend you watch the linked guide.

Heat Set Insert



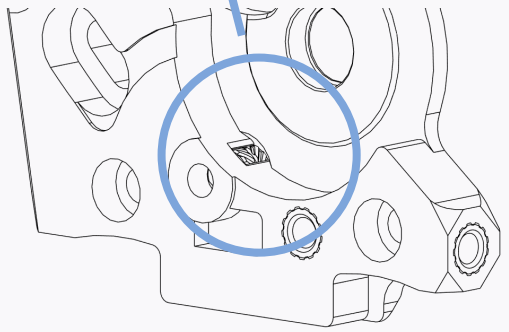
https://voron.link/m5ybt4d





BELOW SURFACE

The top heatset insert needs to sit below the surface of the printed part.



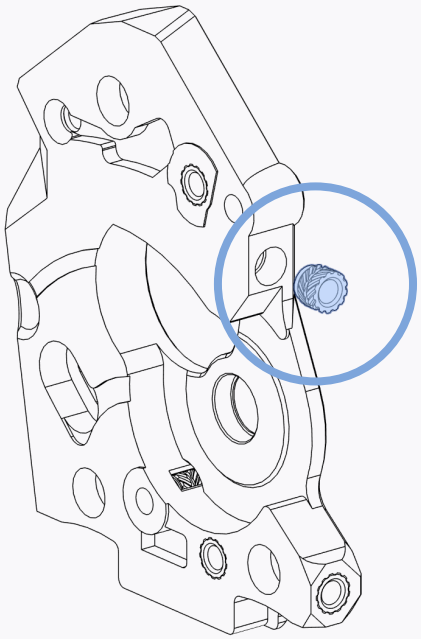
BELOW SURFACE

The bottom heatset insert needs to sit below the surface of the printed part. Mind the cutout in the part and make sure to install it straight.

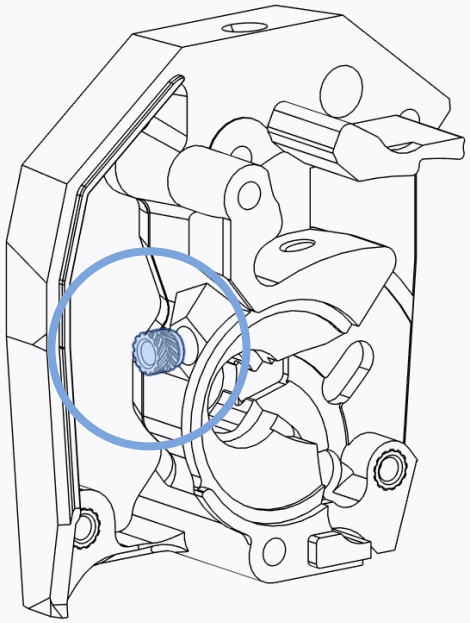
You use a Toolhead PCB, so install these two heat inserts.

OPTION: TOOLHEAD PCB

If you opt to use a toolhead PCB, add additional heat set inserts into locations highlighted below.



Heat Set Insert



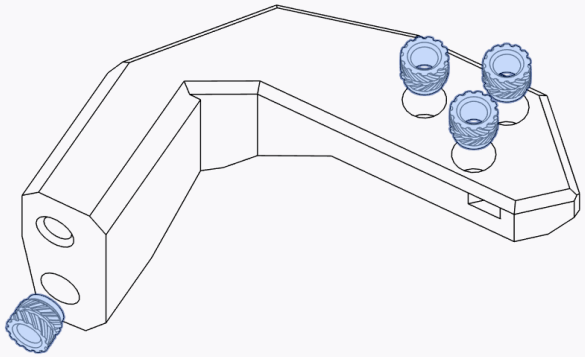
HEAT SET INSERT

Be careful when inserting this heat set insert. It's easy to accidentally touch the left side of the part with the soldering iron.

Skip

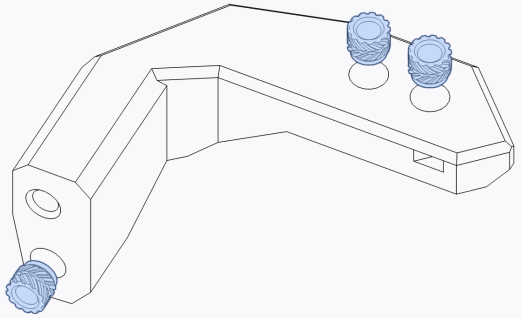
GENERIC CABLE CHAINS

The 3-hole pattern is usually found on generic cable chains.



IGUS CABLE CHAINS

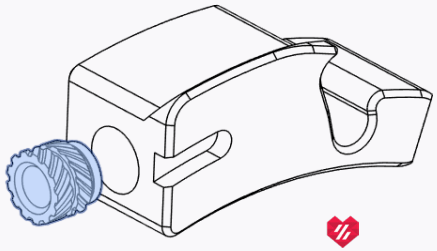
IGUS chains have 2 mounting holes.

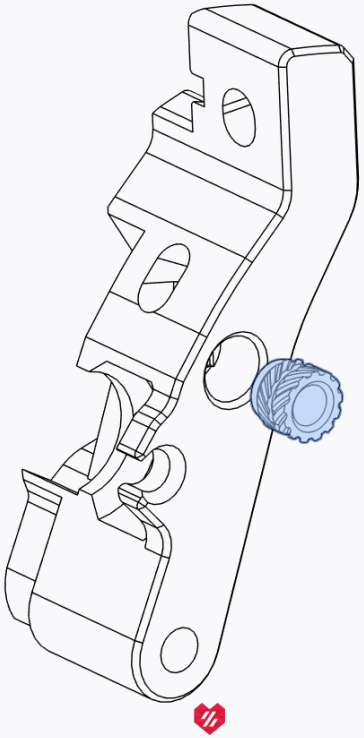


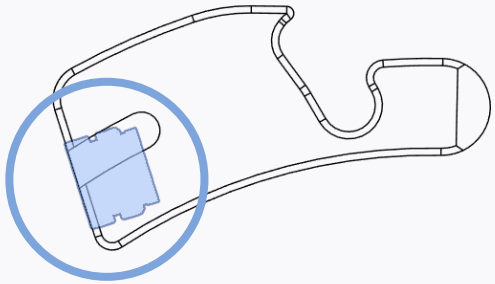
Heat Set Insert

PRINTER SPECIFIC MOUNTS

V2, Trident and Legacy use the same printed parts. Extra parts are included for SwitchWire.





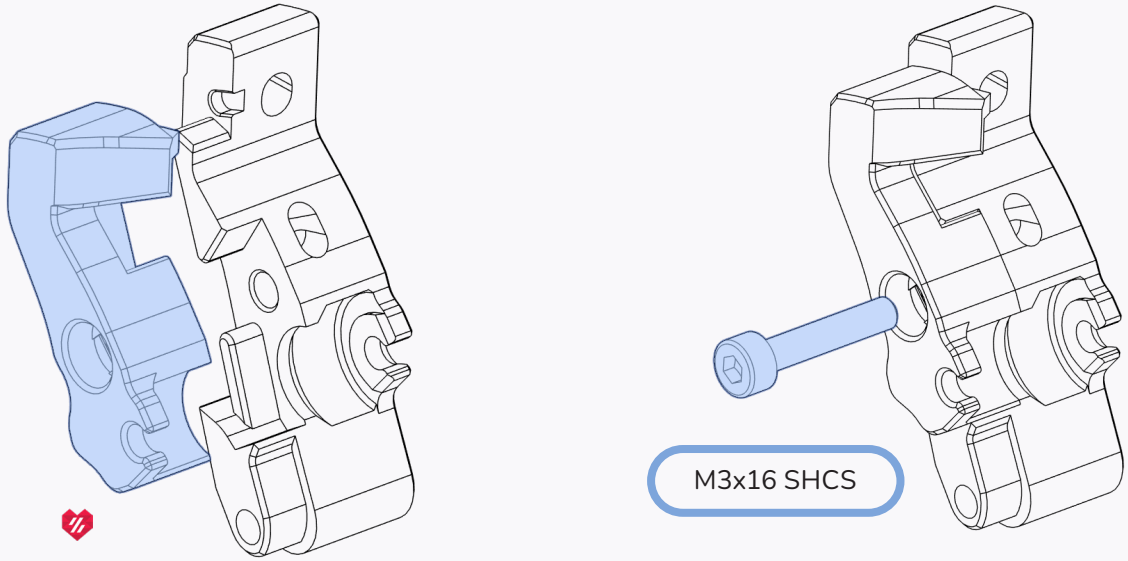


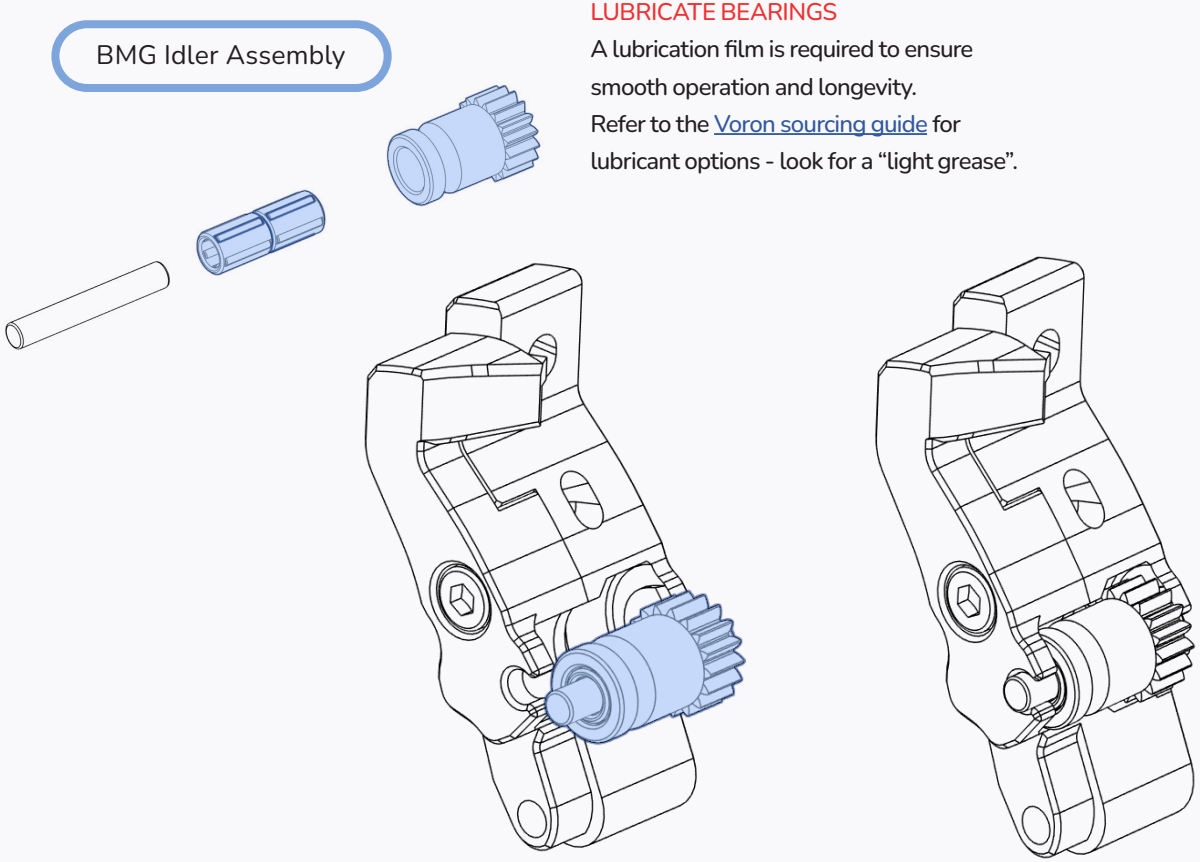
FLUSH WITH SURFACE

The heatset insert needs to sit flush or slightly below the surface of the printed part.

ACCENT PART?

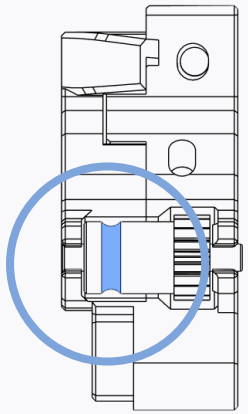
Look for Voron heart next to the part. It indicates that this is a part that is usually printed in the accent color.


M3x16 SHCS



LUBRICATE BEARINGS

A lubrication film is required to ensure smooth operation and longevity. Refer to the Voron sourcing guide for lubricant options - look for a “light grease”.

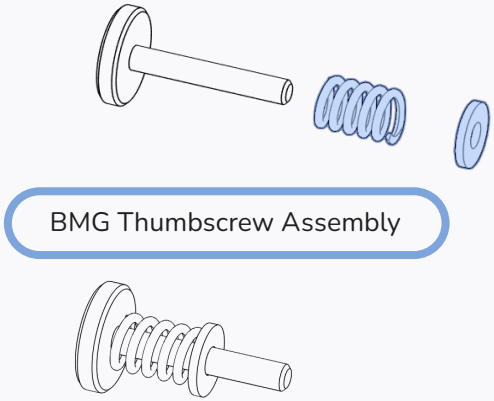


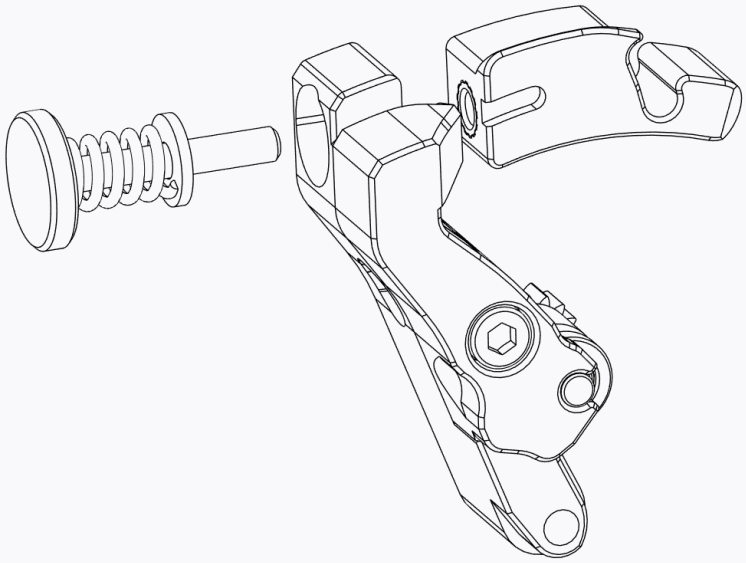
MIND ORIENTATION

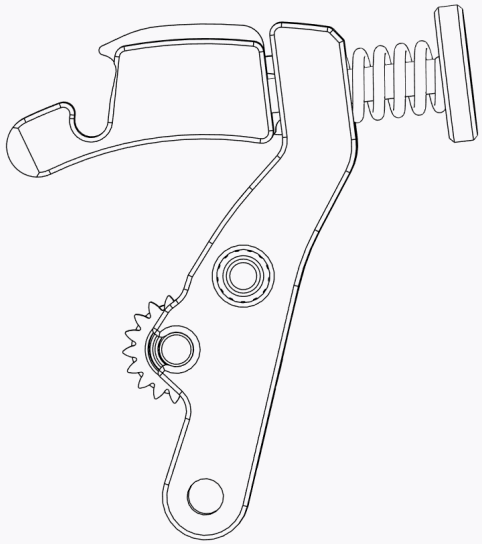
Make sure to orient the idler assembly as shown above.

A NOTE ON SPRINGS

Longer/shorter/stiffer springs will change the tension characteristics and have an impact on how well the tension mechanism works. Consider buying the original Bondtech part as those are known to work well. If sourced from a different vendor check if it's roughly 12mm long with an outer diameter of 6mm and a wire thickness of 1mm.





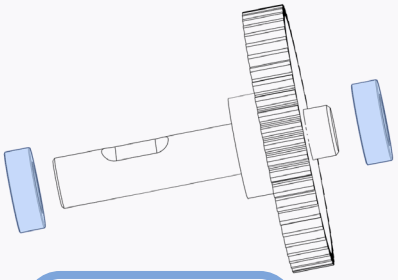


CHECK BEARING FIT

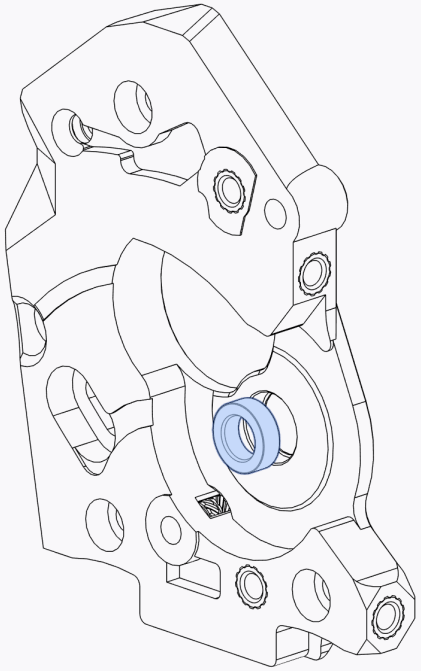
The bearings must slip on and off the shaft easily to allow the gear to self-centre.

Pressing the bearings on the shaft will damage them.

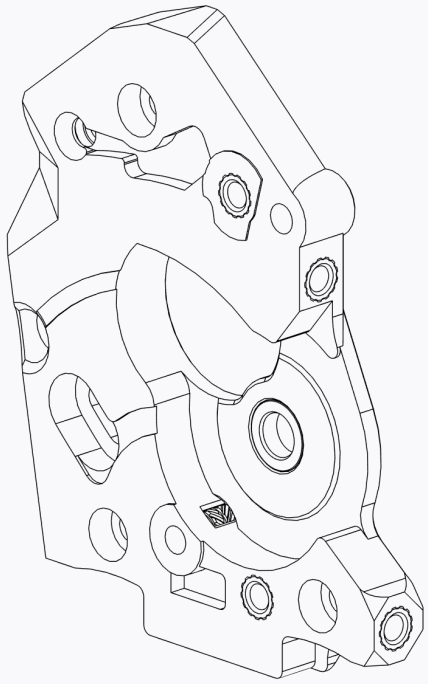
Lightly sand the shaft if required.



MR85 Bearing



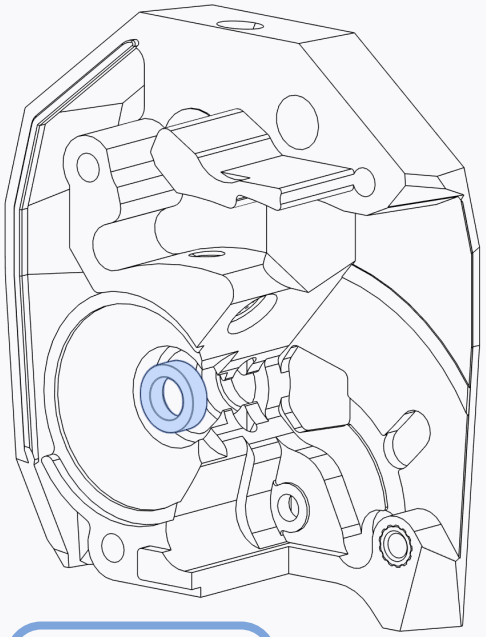
MR85 Bearing



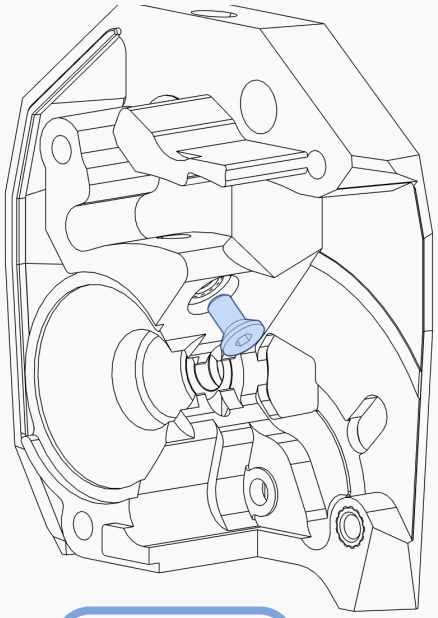
BEARING FIT

Fully seat the bearing into the plastic pocket. Apply even pressure to insert them. Avoid pressing on the inner ring of the bearing.

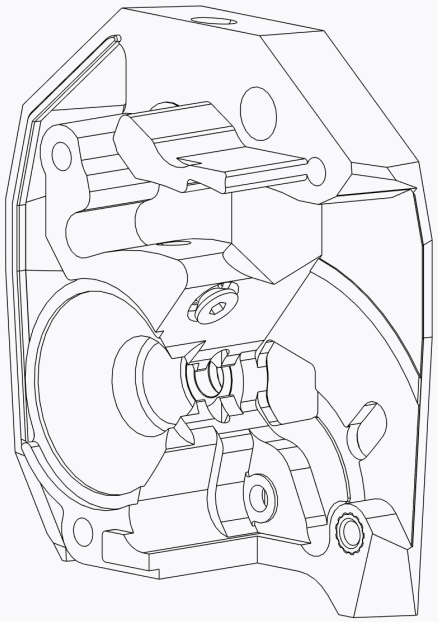
If the fit is too tight the printed parts are likely over-extruded.



MR85 Bearing



M3x6 FHCS



BEARING FIT

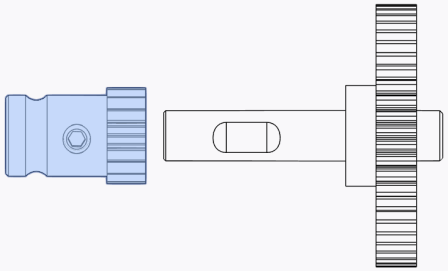
Fully seat the bearing into the plastic pocket. Apply even pressure to insert them. Avoid pressing on the inner ring of the bearing.

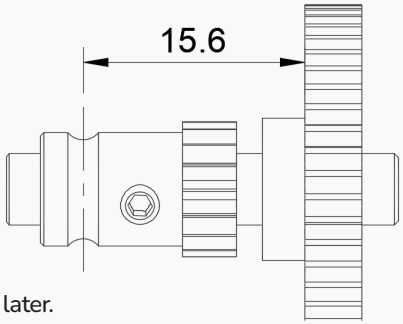
THREAD LOCKER PLEASE

Adding a small amount of medium strength thread locker to this flat head screw will keep it from coming loose over time.

A NOTE ON GEARS

Poorly made gears often cause print quality issues. For best performance consider sourcing the original Bondtech parts.



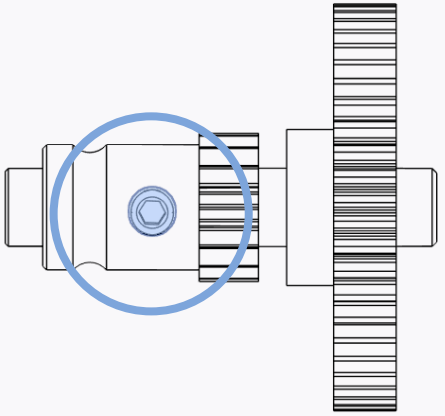


INITIAL POSITION

The final position is set later.

DRIVE GEAR

Make sure the set screw in the filament drive gear is seated against the notch in the shaft. Carefully tighten the set screw, the head is easy to strip.

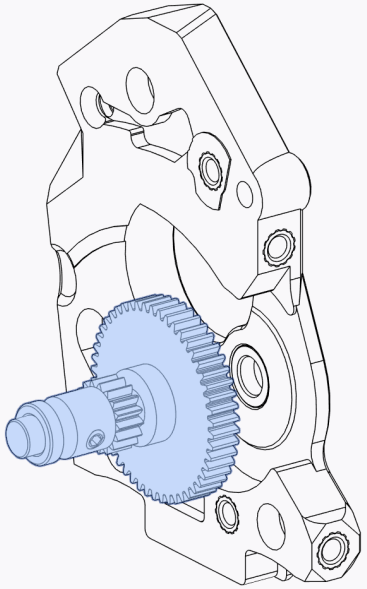


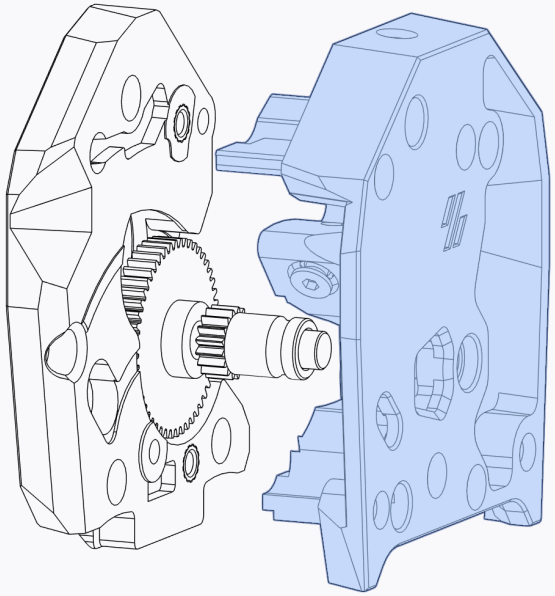
THREAD LOCKER

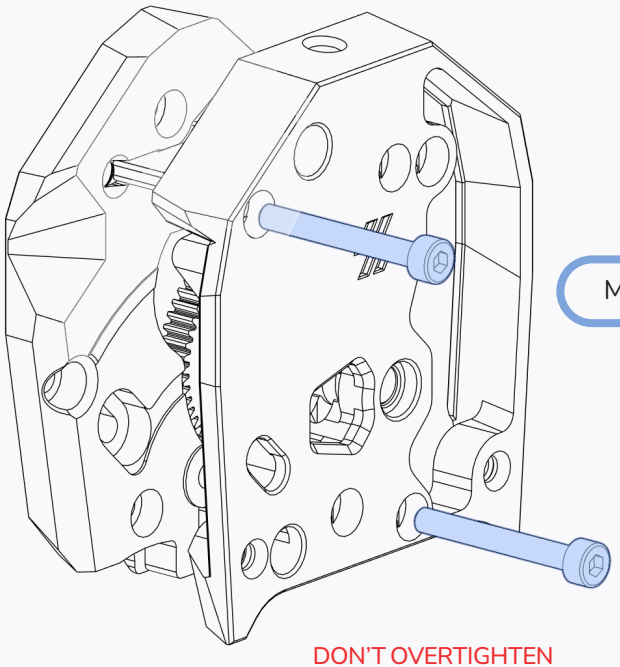
The final position of the drive gear is set in a later step. Common thread lockers have a long enough working time to complete the steps without issues.

Familiarize yourself with the steps on the next 3 pages before you apply thread locker.

Complete the steps on the next 3 pages after applying the thread locker.



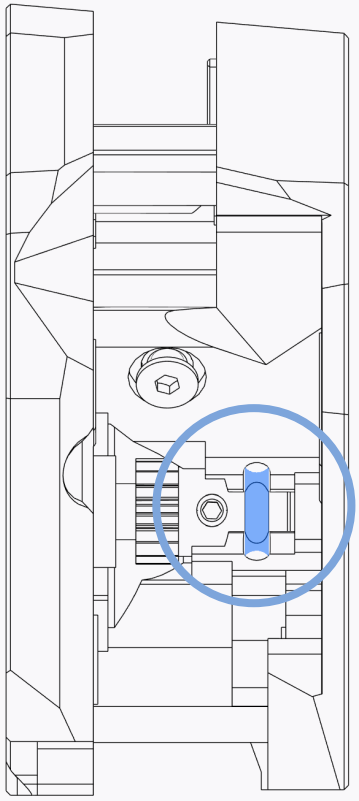




M3x25 SHCS

DON'T OVERTIGHTEN

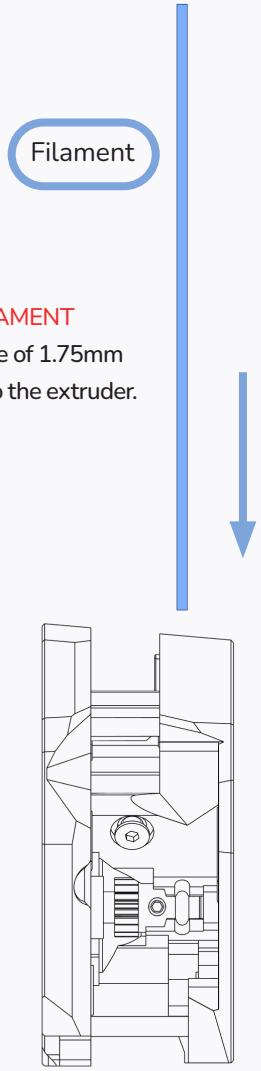
Tighten until the plastic bends and cracks. Back up 2 turns, discard parts, reprint and try again.



INITIAL ALIGNMENT CHECK

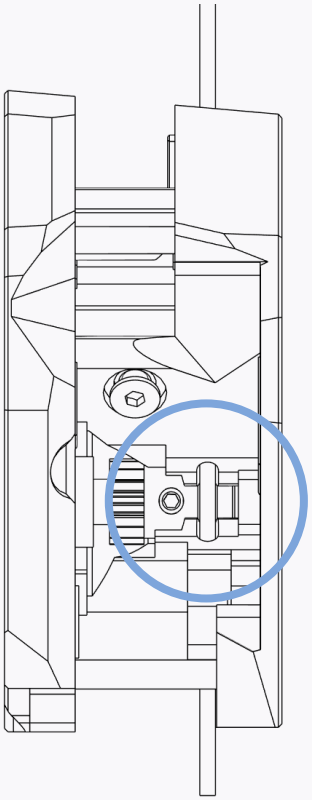
Check if the filament path aligns with the toothed section of the drive gear.

DRIVE ALIGNMENT



INSERT FILAMENT

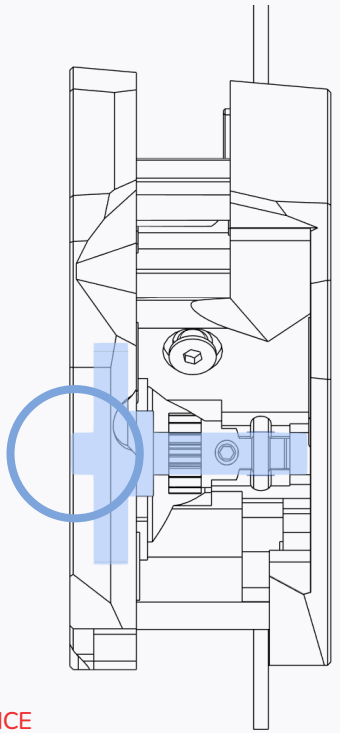
Insert a piece of 1.75mm filament into the extruder.



CHECK ALIGNMENT

With the filament inserted, verify if the filament path and drive gear are aligned.

Loosen the set screw and adjust the position of the drive gear if required.

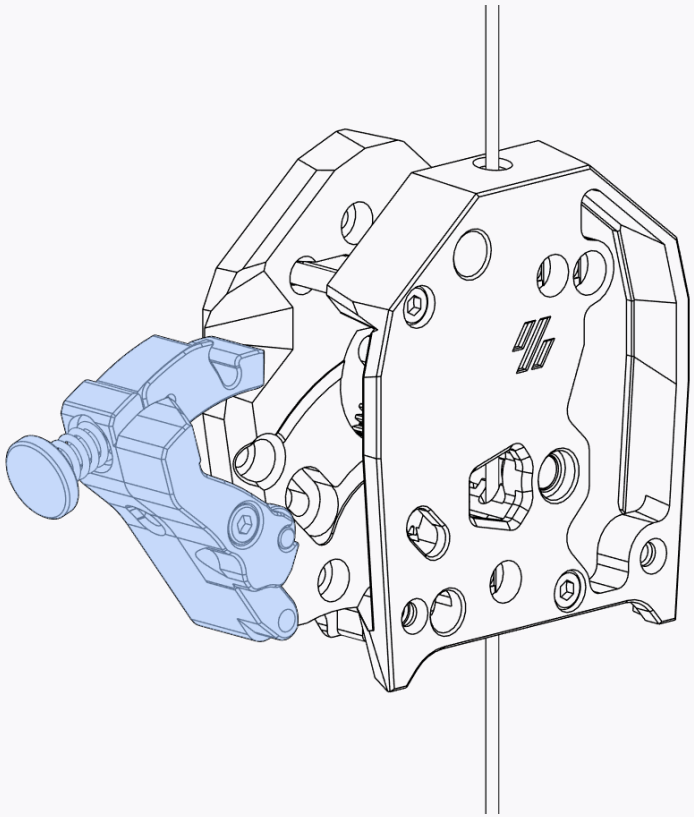


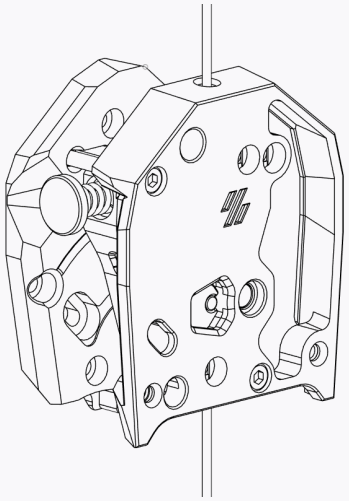
CHECK FOR CLEARANCE

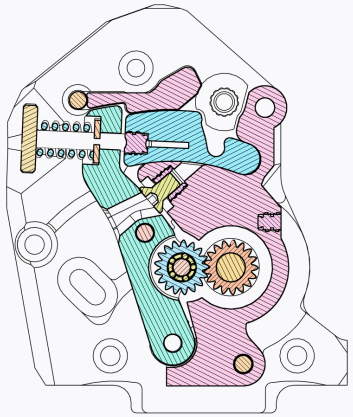
The drive shaft must not touch the motor housing. Make sure it does not sit above the surface of the printed part.

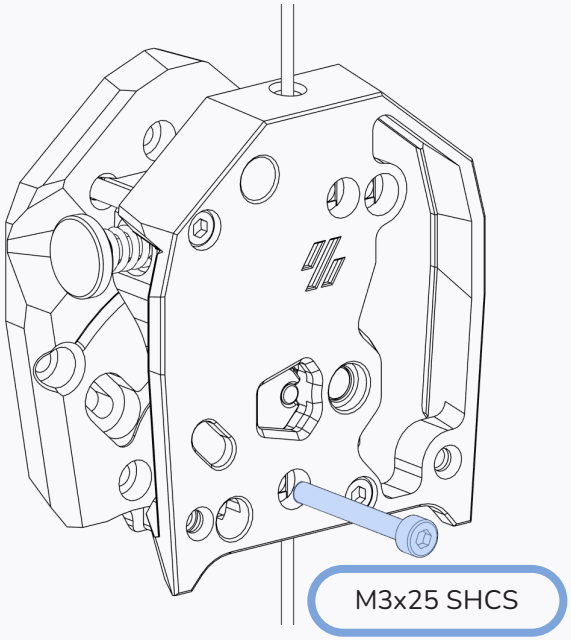
Check if the shaft has sufficient clearance when fully seated.

Depending on the shafts tolerances you may need to adjust the position of the drive gear or sand the face of shaft.



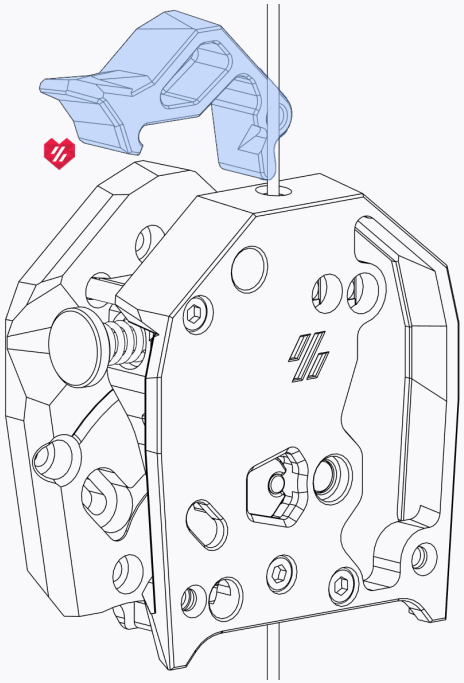


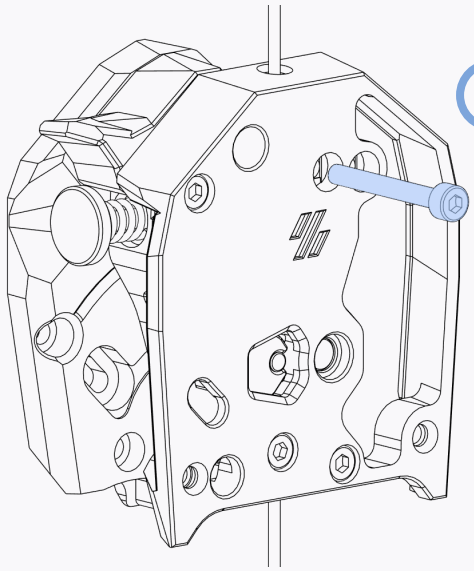


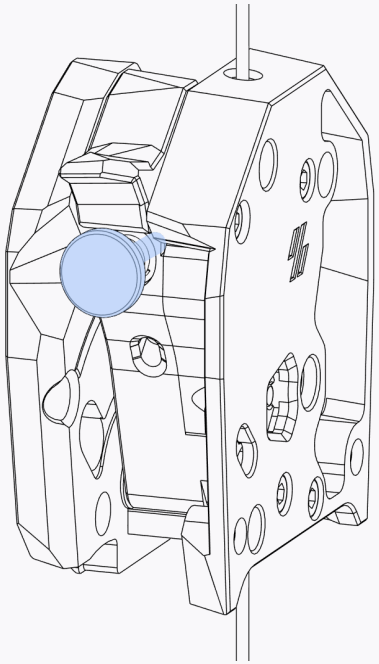


DON'T TIGHTEN

The tensioning arm must move freely. Tightening the bolt may lock it in place.

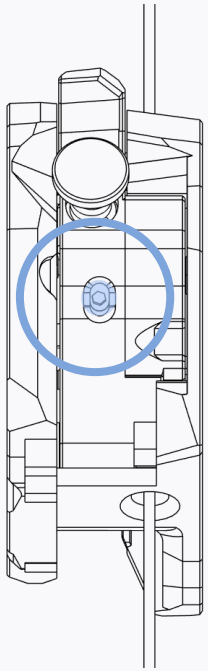



M3x25 SHCS



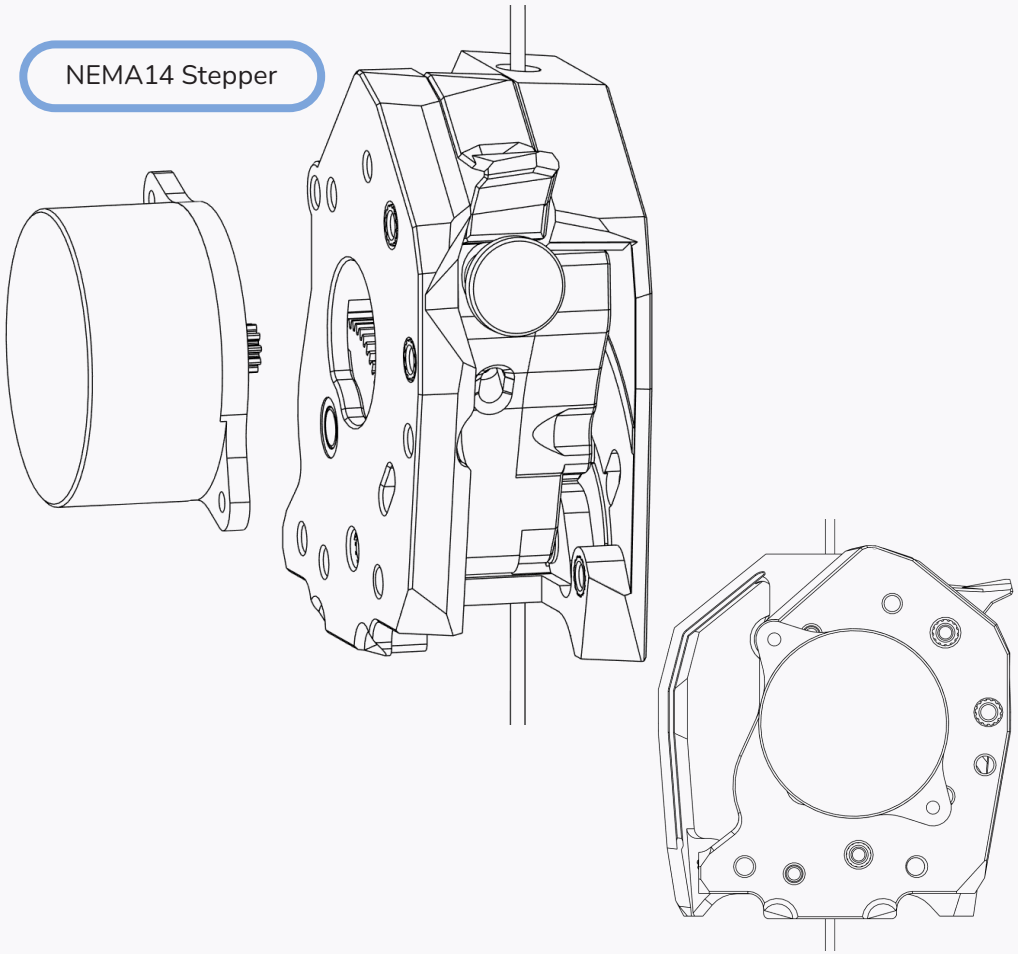
TENSION KNOB

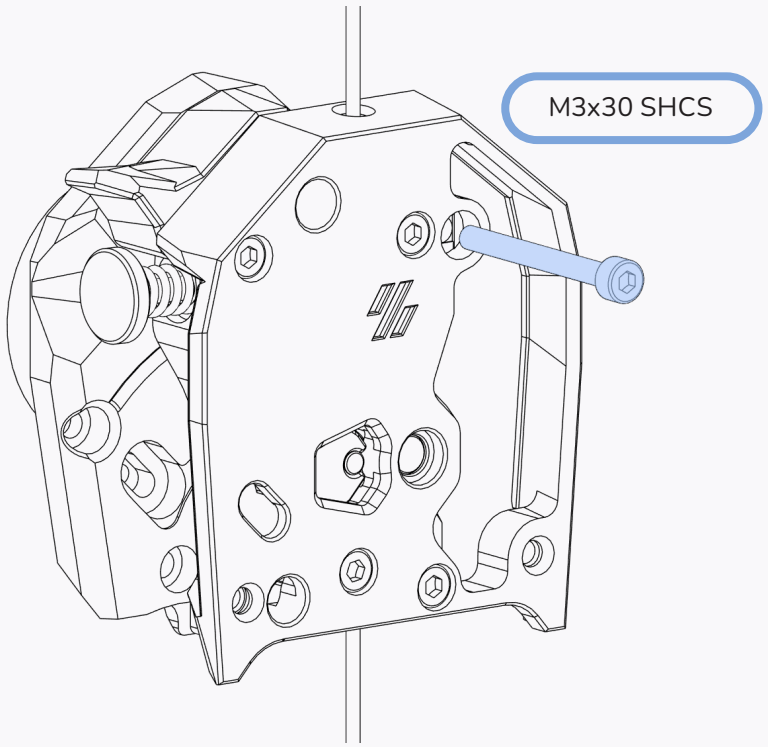
Turning the thumb screw clockwise will increase the tension and grip on the filament. Too much tension will result in print issues.

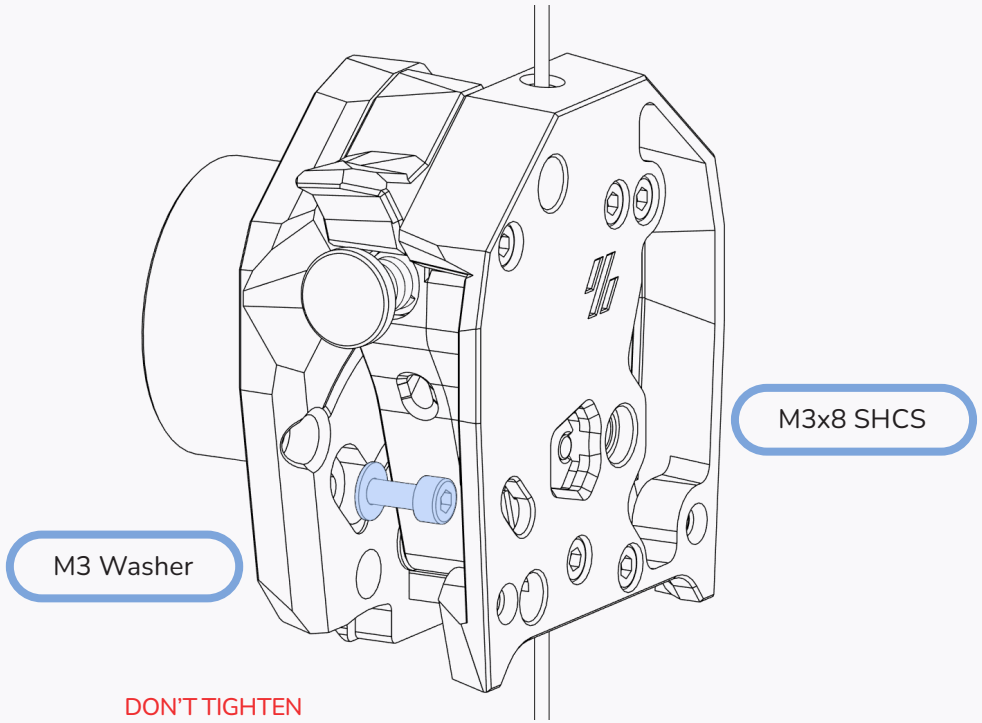


ANTI SQUISH THINGYMAJIG

Softer and flexible materials will deform and extrude poorly under too much tension. ClockWork2 adds an adjustment feature to set the minimum distance between the drive gear and the idler, limiting the squish on the filament, and to prevent the gears from meshing too tightly or binding up the extruder.

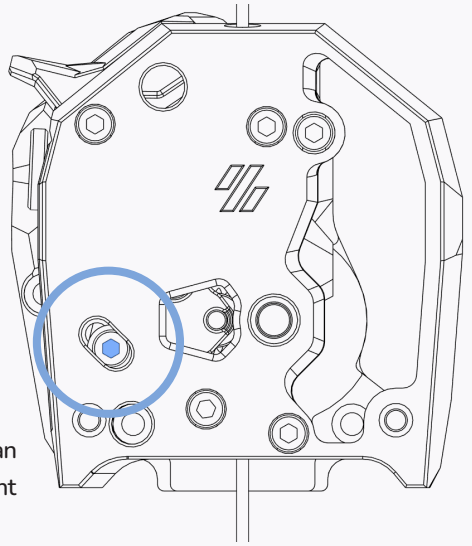

NEMA14 Stepper


M3x30 SHCS



DON'T TIGHTEN

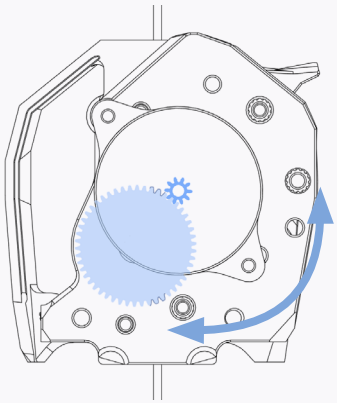
The motor position will be adjusted in the next steps. Don't fully tighten just yet.



ACCESS HOLE

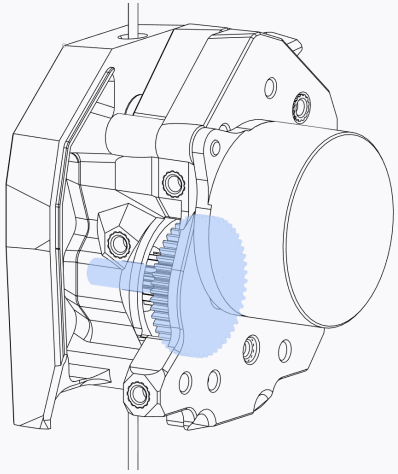
The second motor bolt can be accessed from the front of the extruder.

Follow this Video Guide on how to set the gear mesh on your CW2. Follow this video Guide on how to set the Anti-Squish as well.



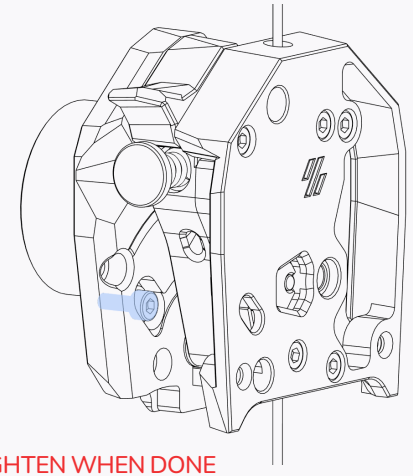
SET GEAR MESHING

Adjust the stepper motor position so that the motor and extruder gear teeth fully mesh /overlap with each other. There should be a very small gap between the faces of the gear teeth; the gears must not press tightly against each other.



CHECK GEAR PLAY

The gear should have a slight play and should not be fully tight against the pinion. Adjust the position of the motor until you have a faint play.

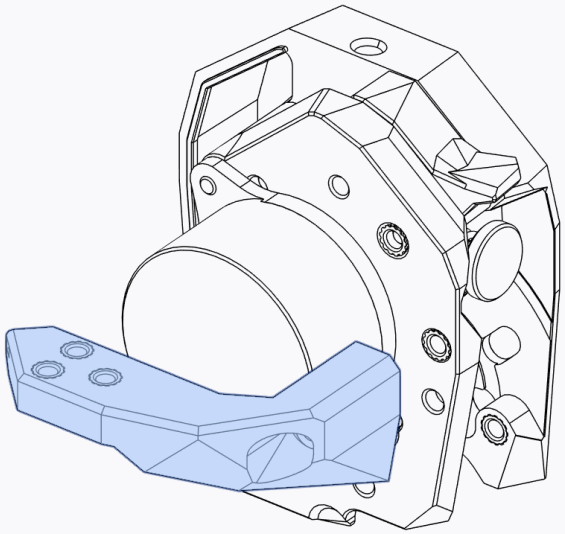


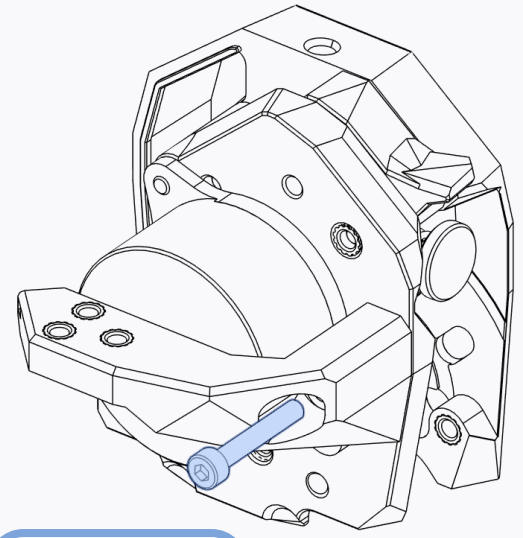
TIGHTEN WHEN DONE

Don't forget to tighten the second motor bolt after adjusting.

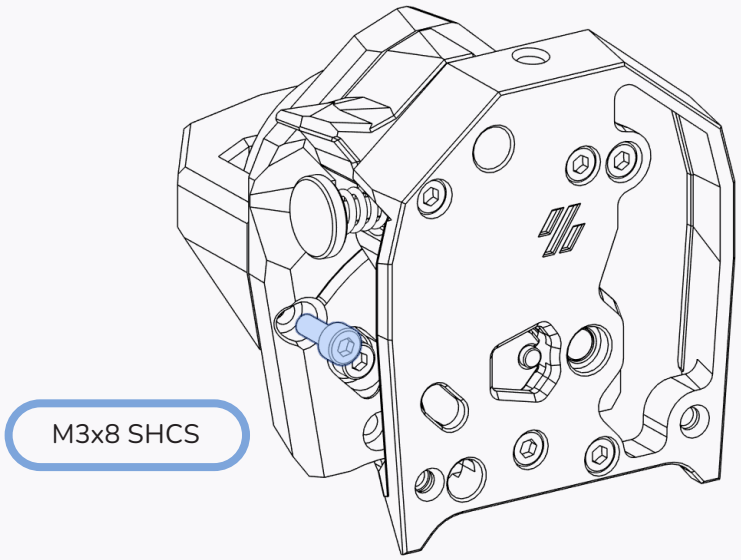
https://voron.link/u6ehxsw

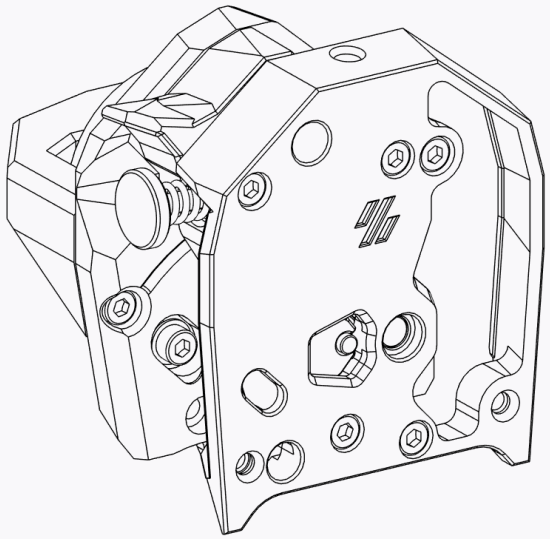
Skip for now, we will install our Version of the Anchor later.

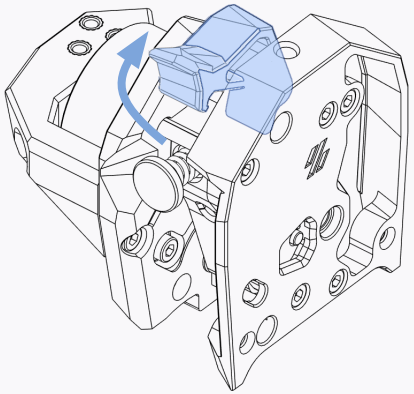




M3x20 SHCS

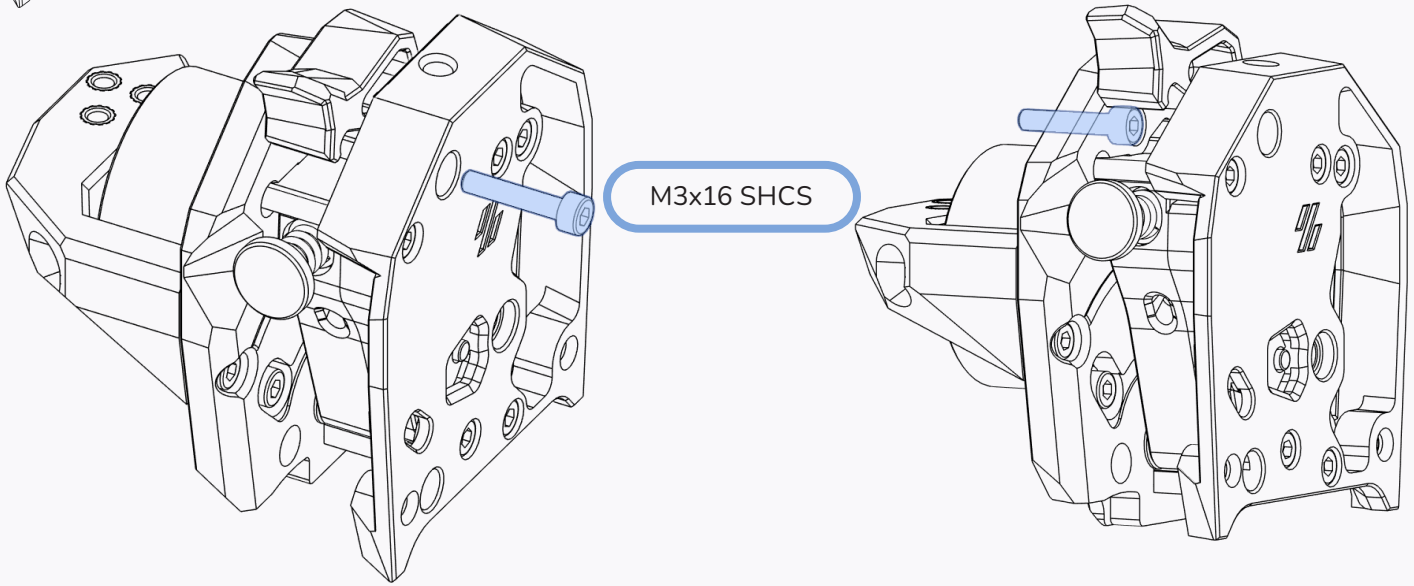

M3x8 SHCS



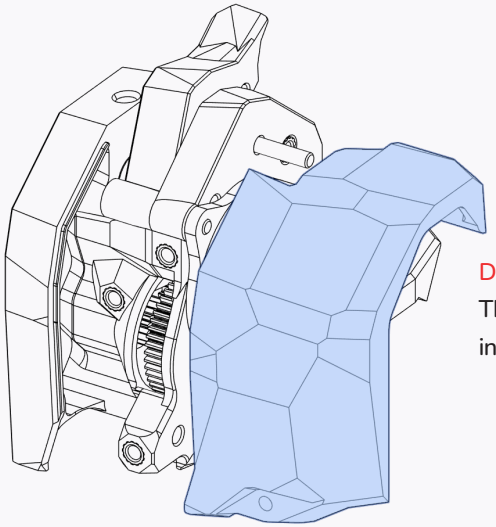


OPEN LATCH

Undo the filament latch to expose the bolt pocket for the cable cover.

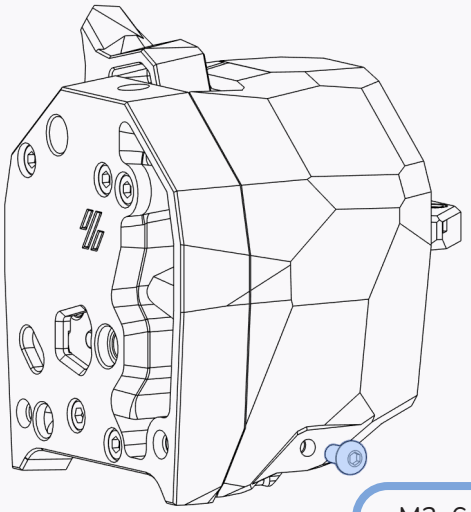


You dont need to fix the Cable Cover down with the m3x6 BHCS now, you will need to access this later.



DON'T OVER-TIGHTEN

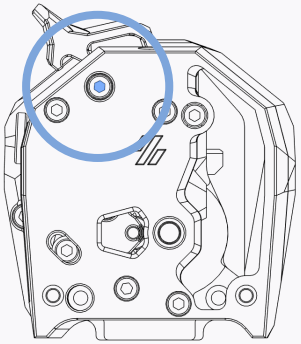
The bolt is threaded directly into plastic.



M3x6 BHCS

ACCESS HOLE

The bolt drive can be accessed from the front of the extruder.



StealthBurner development started on 2021-04-13 with a simple: “I’m going to regret saying this but, ‘How hard could it be?’”

All Hotends are mounted in a similar pattern and pretty self-explaining, your hotend mount and hotend will look different.

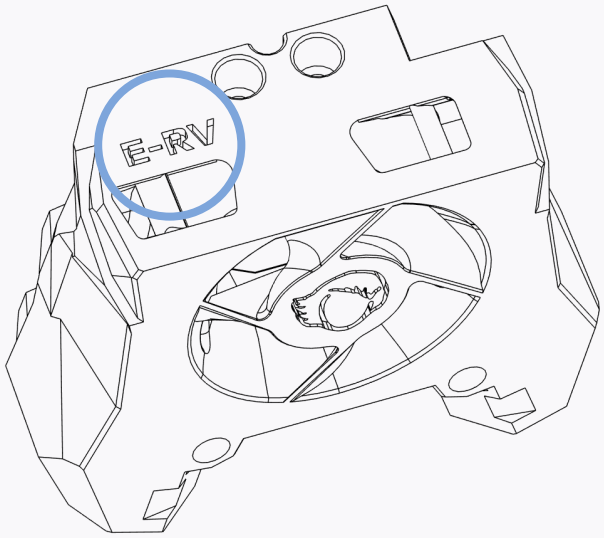


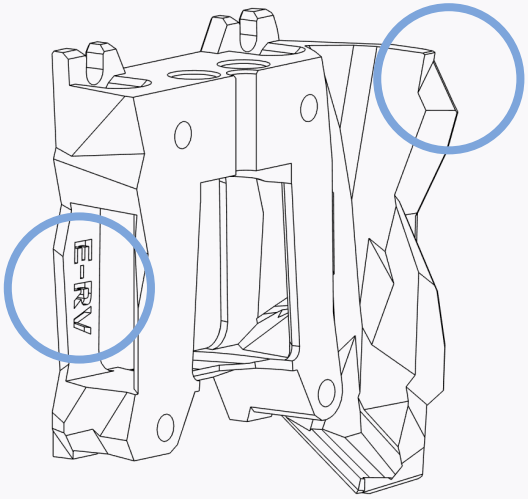
All Hotends are mounted in a similar pattern and pretty self-explaining, your hotend mount and hotend will look different.

IDENTIFYING PRINTED PARTS

The hotend type is embossed on the printed parts.

Make sure they match your hotend.





MISSING CORNER? CW1? CW2?

We also provide tool cartridge parts for ClockWork1, and other extruders designed as an alternative to CW1. They are identical except for an opening in the highlighted corner to help with wire routing for those designs.

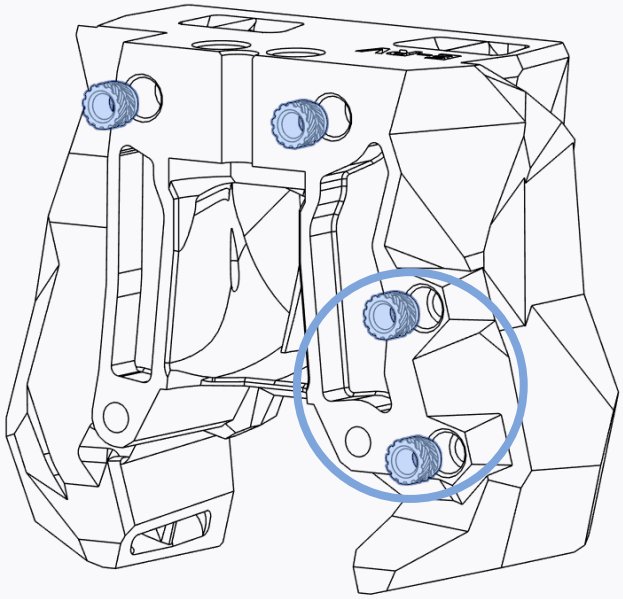
AVAILABLE MOUNTS

We also provide mounts for other hotends. They are assembled in a similar manner.

For a full list of available mounts and their identification code, see the readme file included in the toolheads folder.

All Hotends are mounted in a similar pattern and pretty self-explaining, your hotend mount and hotend will look different.

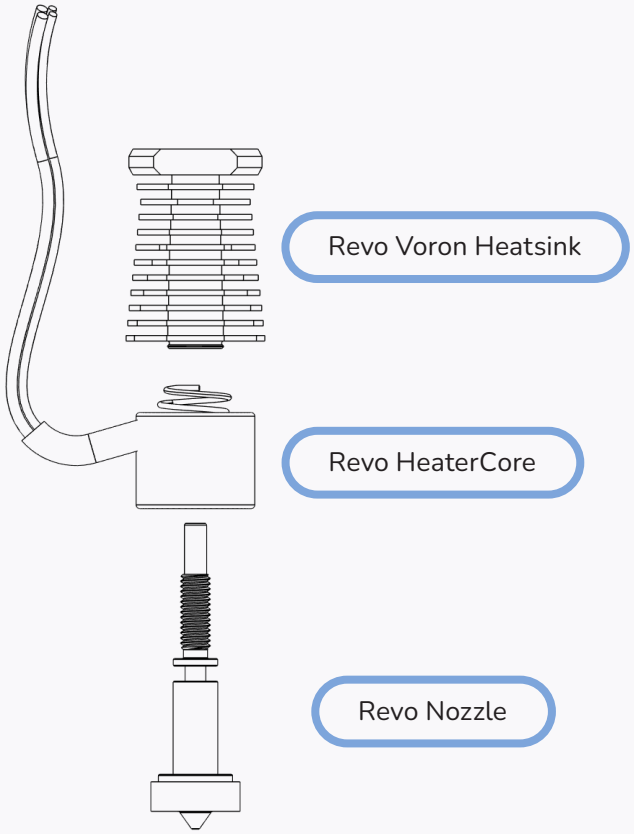
Heat Set Insert

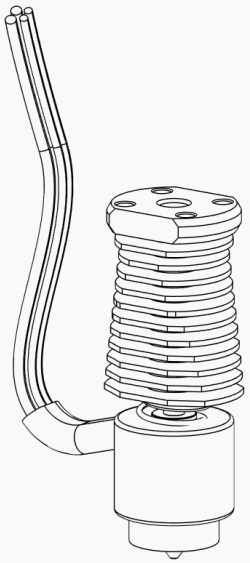


OPTION: ADXL PCB MOUNT

To use the optional ADXL PCB for Klipper's Input Shaper calibration, add additional heat set inserts into the holes in highlighted location.

All Hotends are mounted in a similar pattern and pretty self-explaining, your hotend mount and hotend will look different.





All Hotends are mounted in a similar pattern and pretty self-explaining, your hotend mount and hotend will look different.

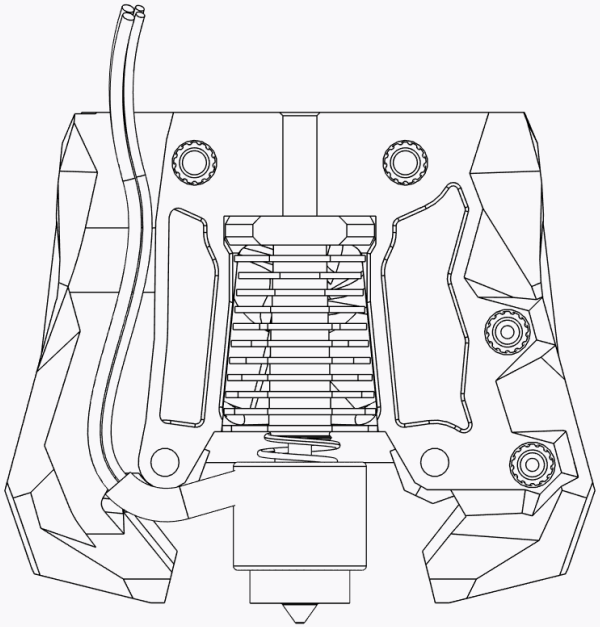
HOTEND WIRE ROUTING

Route the wires as shown to the right. This is universal to all hotend types.

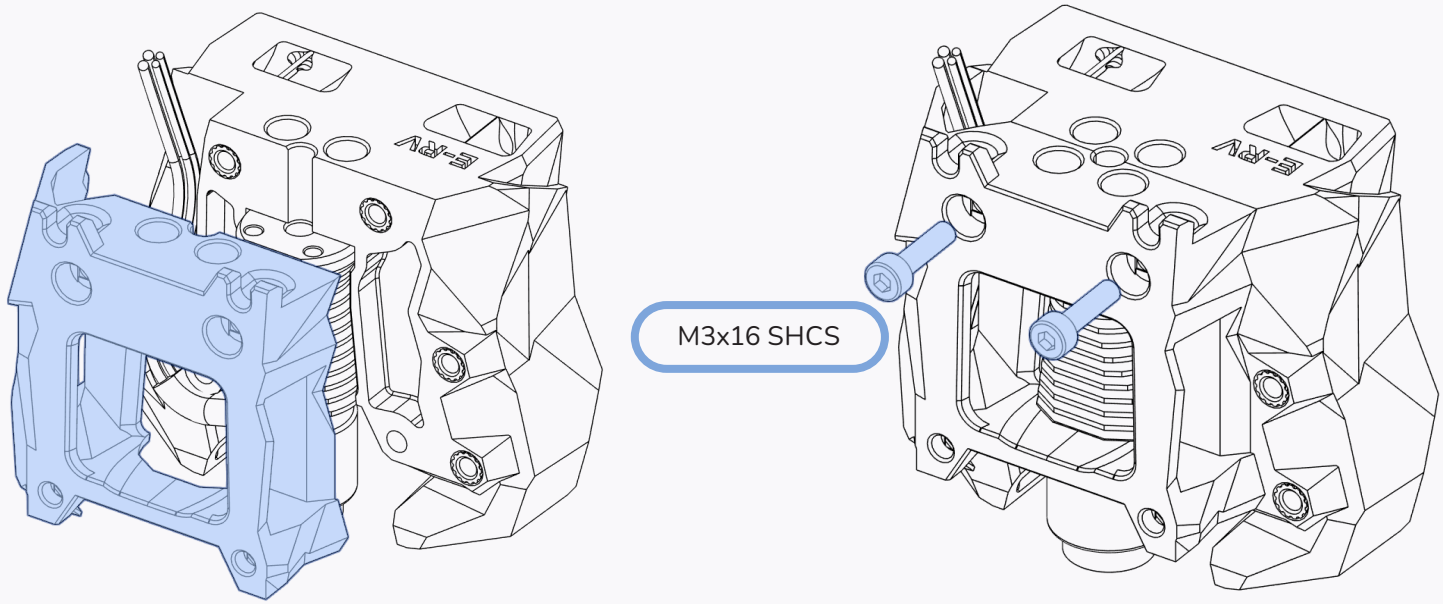
BEND STRAIN RELIEF

Carefully bend the strain relief to clear the printed part. Firmly hold the HeaterCore to prevent bending the Revo Nozzle.

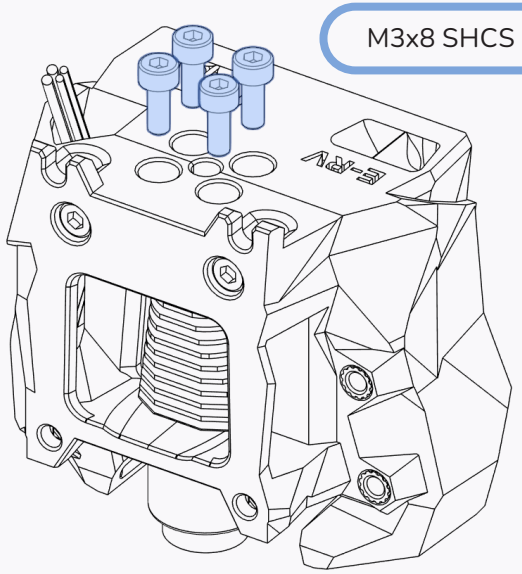
It is critical that the Revo Nozzle is fully hand tightened into the heatsink. You MUST bend the strain relief to achieve this.



All Hotends are mounted in a similar pattern and pretty self-explaining, your hotend mount and hotend will look different.



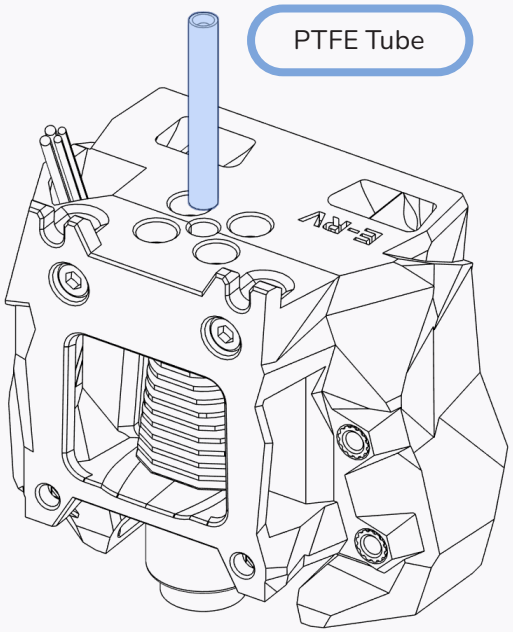
All Hotends are mounted in a similar pattern and pretty self-explaining, your hotend mount and hotend will look different.



M3 MOUNTING BOLTS?

Cause we can, that's why.

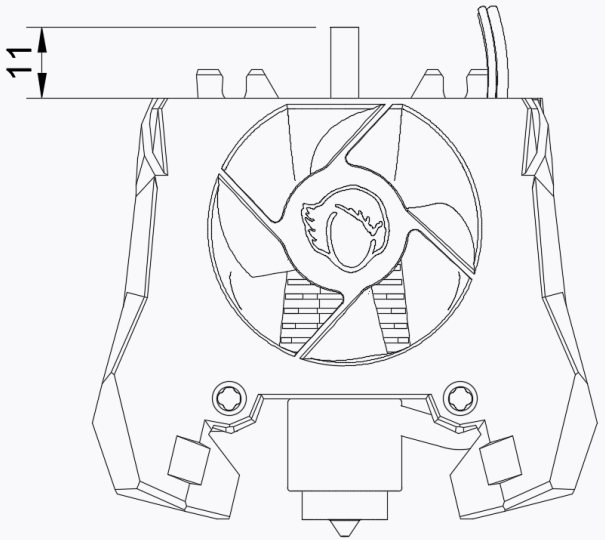
All Hotends are mounted in a similar pattern and pretty self-explaining, your hotend mount and hotend will look different.

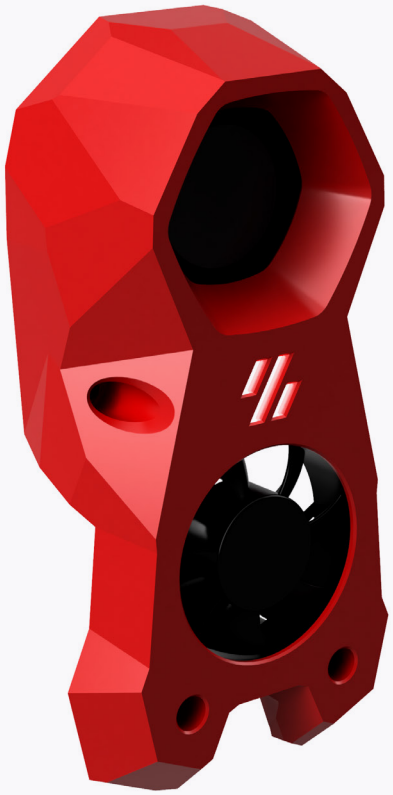


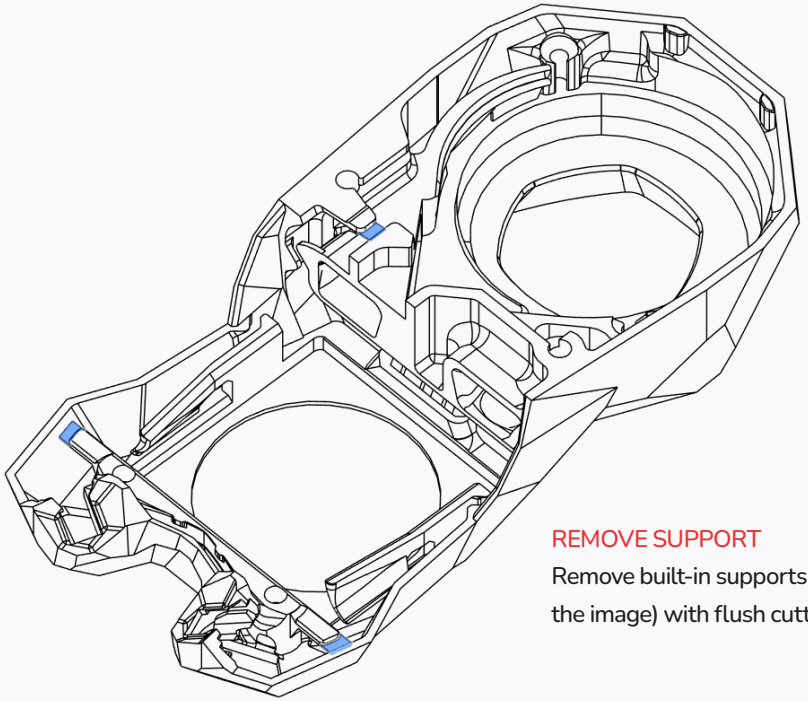
PTFE STICKOUT

The PTFE tube should stick out 11mm above the surface of the printed part.

The stick out length might vary if you use an extruder other than the ClockWork2.



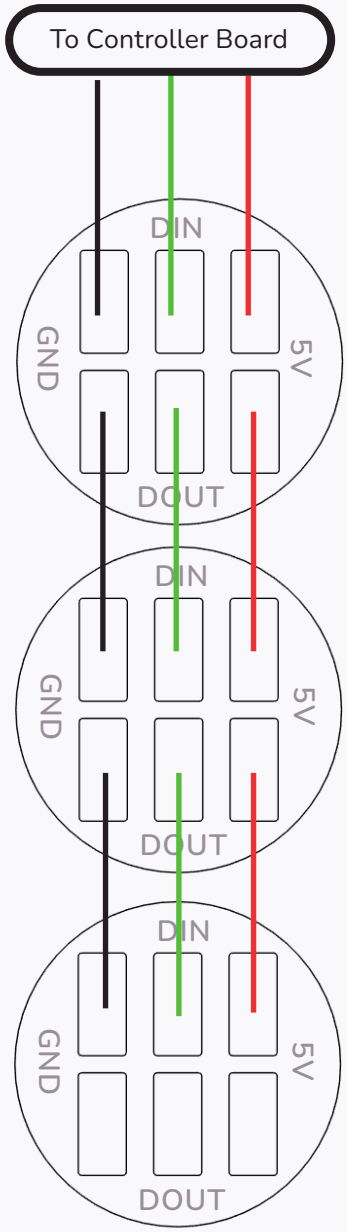




REMOVE SUPPORT

Remove built-in supports (highlighted in the image) with flush cutters.

Be very careful with the cables, you can easily rip them off the LED PCB which leads to this LED and the following ones not working.



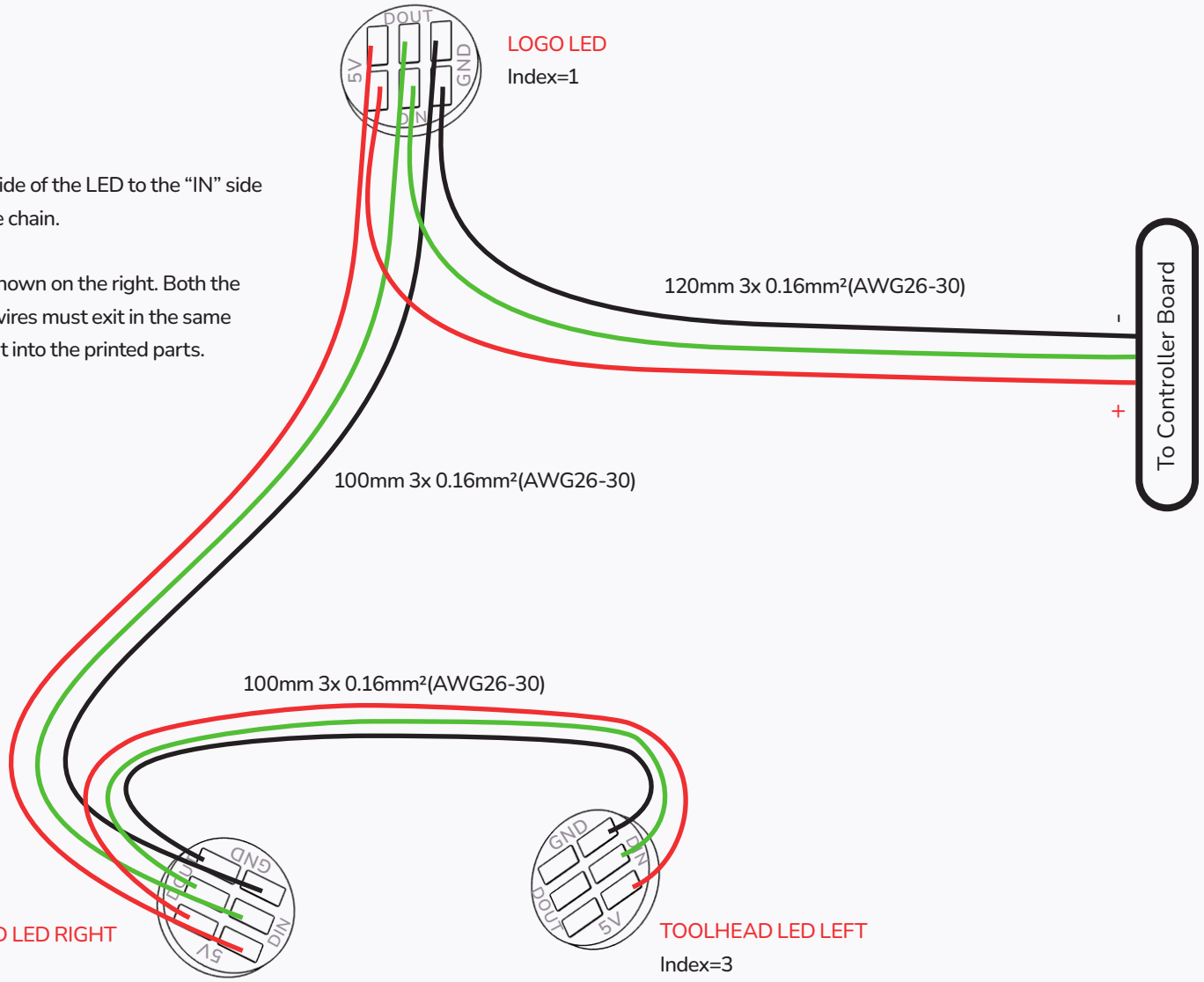
For Illustration Only, Read "LED Wiring" Callout

VORONDESIGN.COM

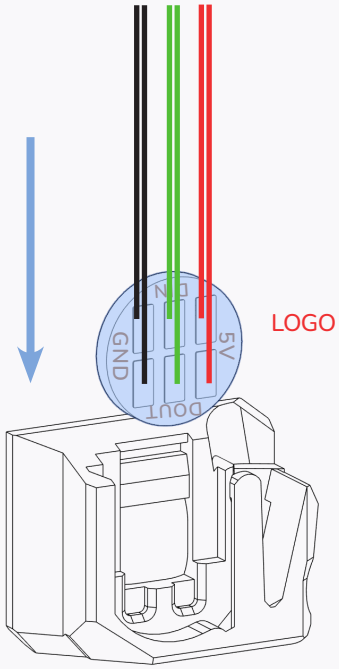
LED WIRING

Connect the "OUT" side of the LED to the "IN" side of the next LED in the chain.

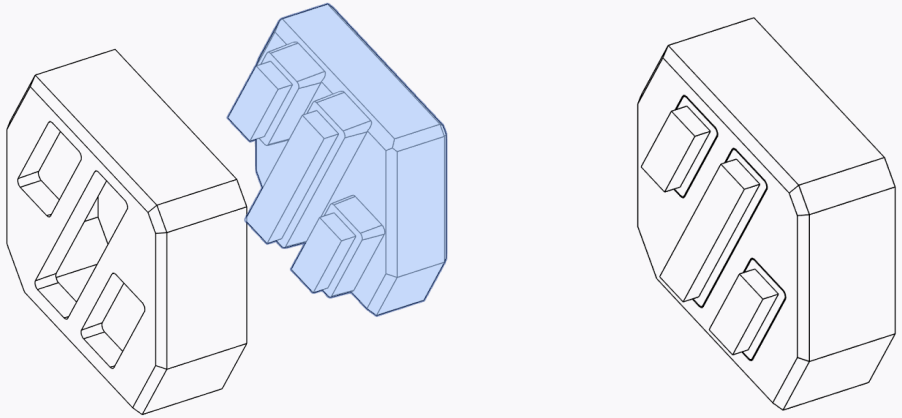
Solder the wires as shown on the right. Both the "IN" and the "OUT" wires must exit in the same direction in order to fit into the printed parts.



Be very careful with the cables, you can easily rip them off the LED PCB which leads to this LED and the following ones not working.



LOGO LED



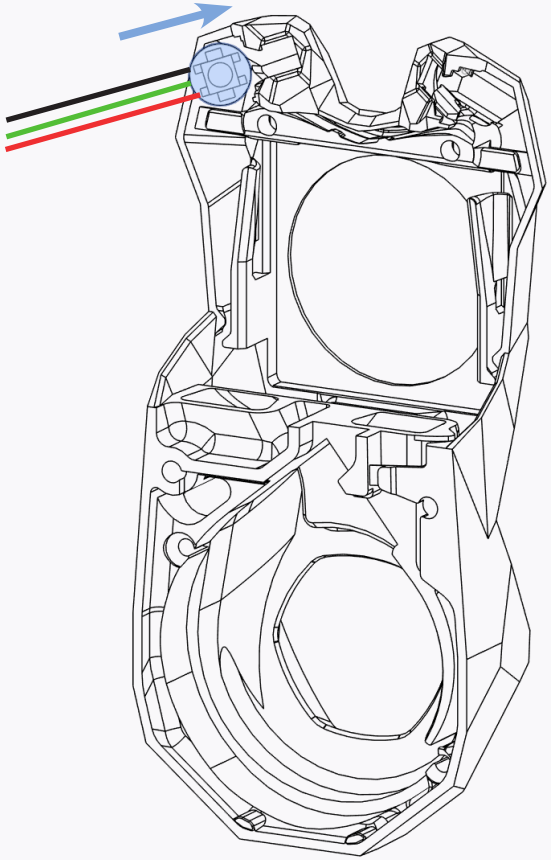
LED DIFFUSER AND MASK

The diffuser (highlighted part) is printed in a translucent filament to evenly spread the light.

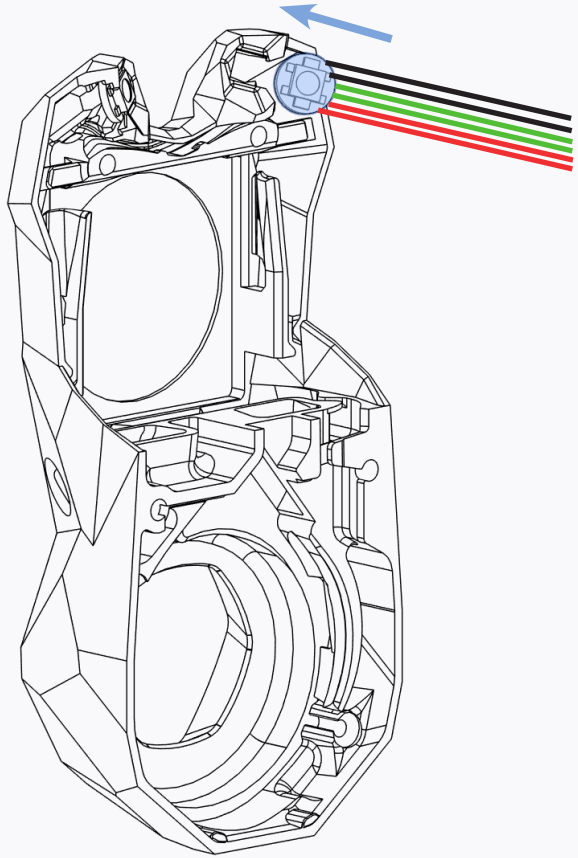
The mask (part to the left) is printed in an opaque filament to prevent the light from leaking, resulting in a crisp logo.

Be very careful with the cables, you can easily rip them off the LED PCB which leads to this LED and the following ones not working.

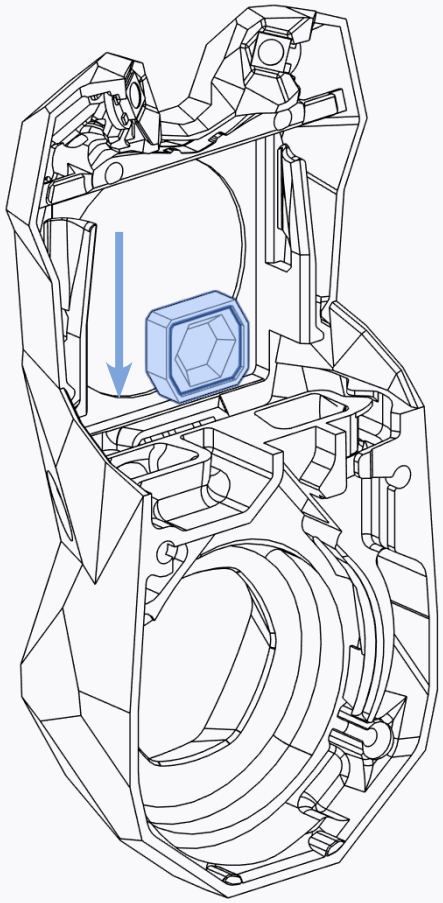
TOOLHEAD LED LEFT



TOOLHEAD LED RIGHT

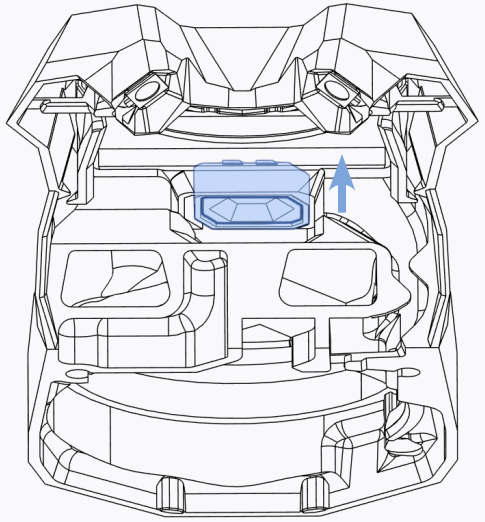


Be very careful with the cables, you can easily rip them off the LED PCB which leads to this LED and the following ones not working.

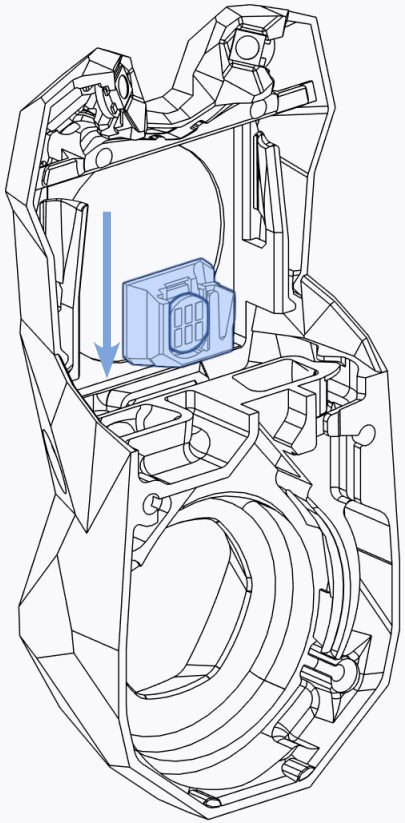


DIFFUSER INSERTION

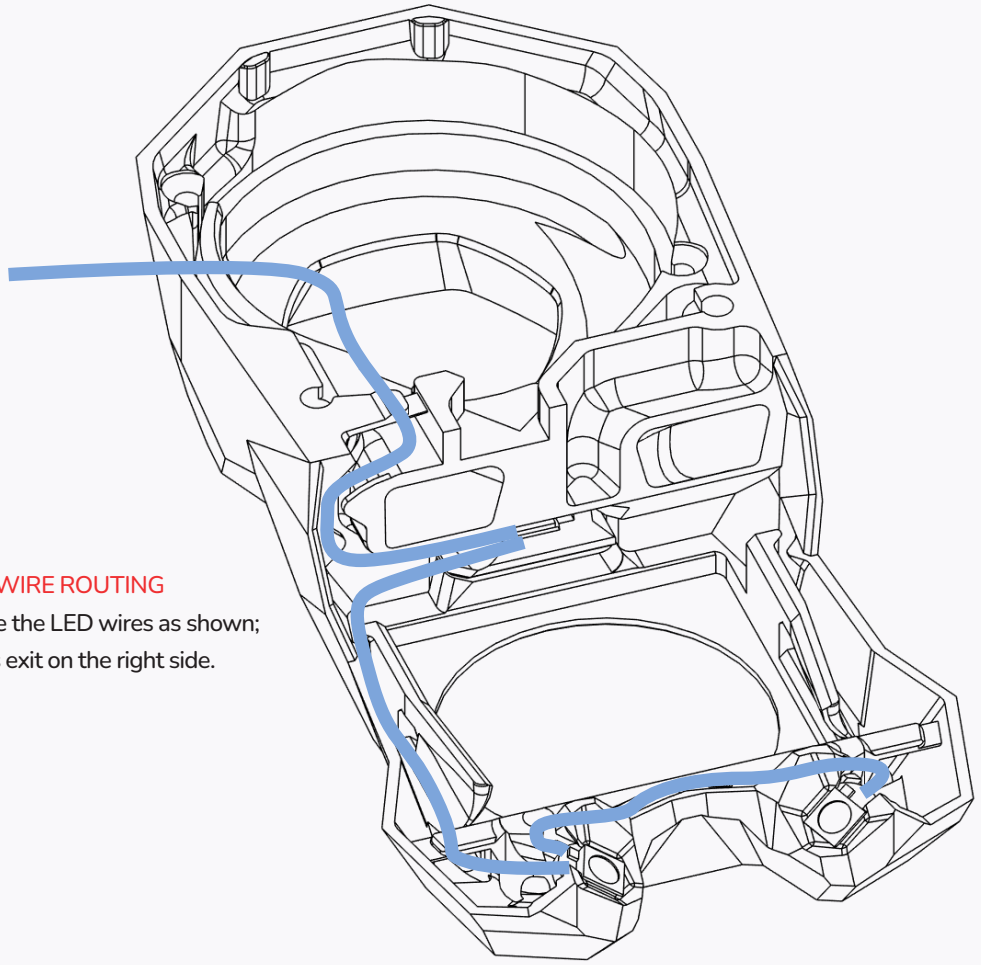
Insert the printed parts and push them towards the front.



Be very careful with the cables, you can easily rip them off the LED PCB which leads to this LED and the following ones not working.

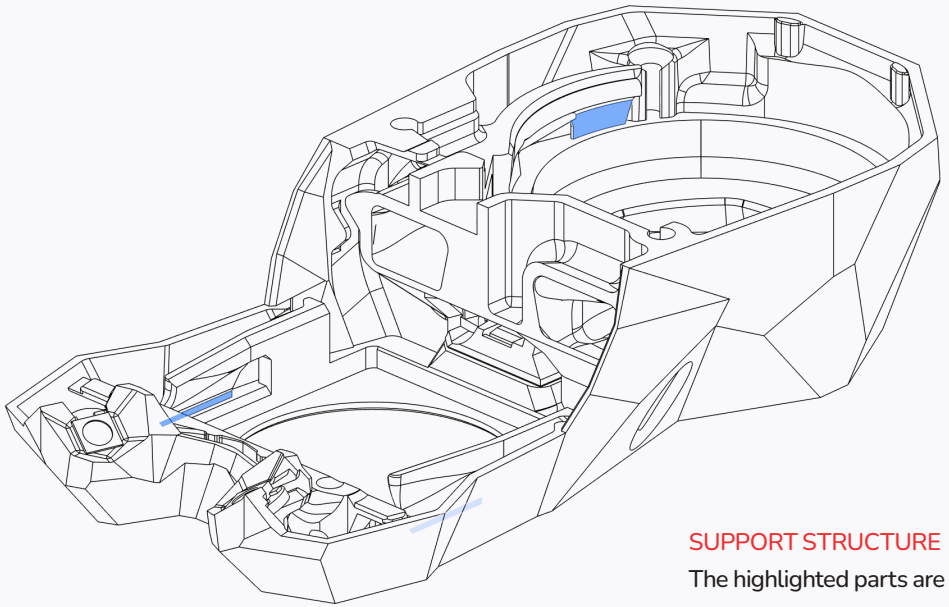


Be very careful with the cables, you can easily rip them off the LED PCB which leads to this LED and the following ones not working.



LED WIRE ROUTING

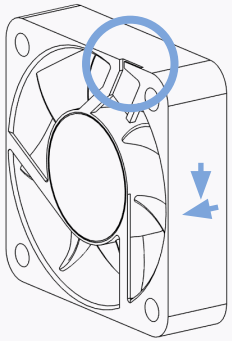
Route the LED wires as shown; wires exit on the right side.



SUPPORT STRUCTURE

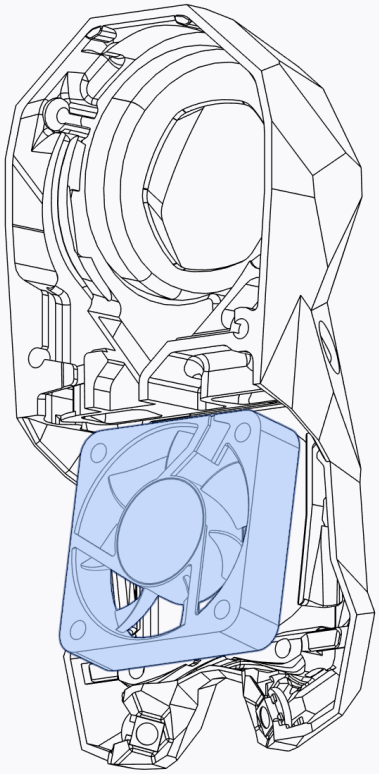
The highlighted parts are build-in support structures. They are designed to break during the fan installation.

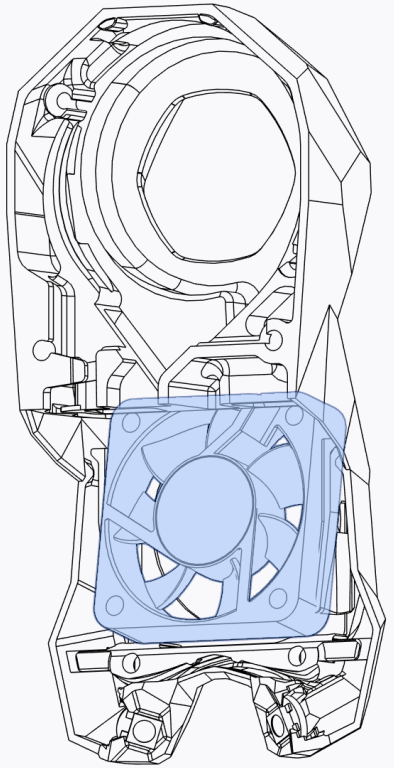
Be sure that your fan blows in the right Direction. Look at the arrows on the fan. Additionally, the Sticker on the fan is on the back pointing towards the Hotend. You want the fan to blow air into the hotend, not trying to pull the air out. This destroys your fan and your whole hotend mount because of heatcreep.



FAN ORIENTATION

Rotate the fan so that the wires exit on the top and the air is pushed “inwards”.





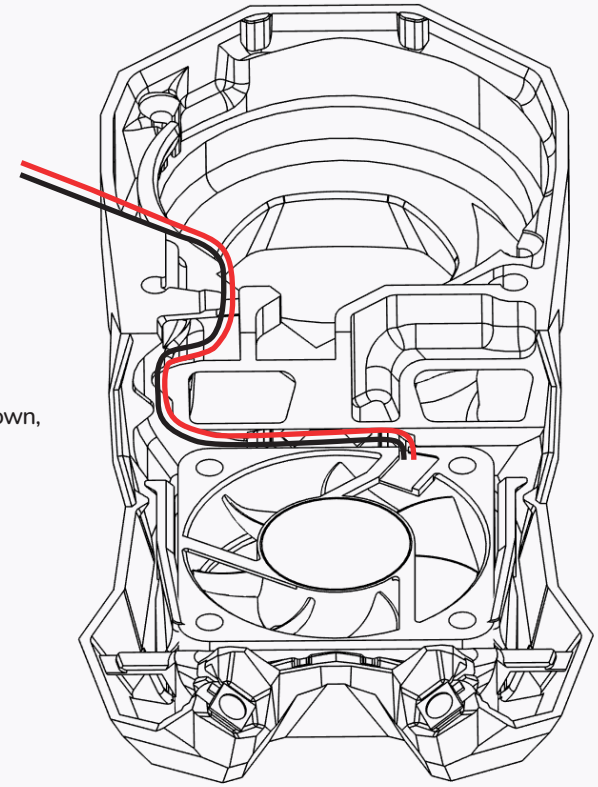
FAN INSERTION

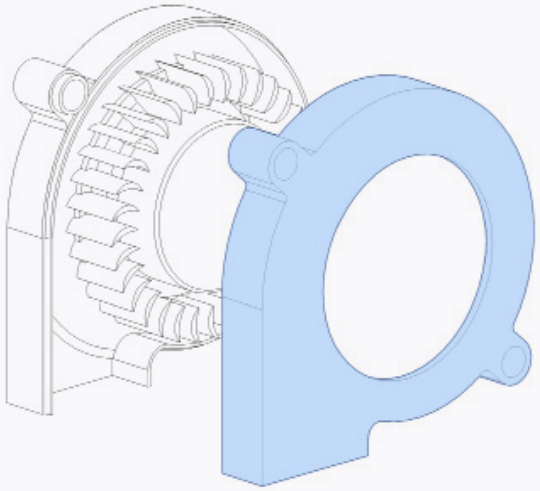
Insert the fan at a slight angle and clip it into place.

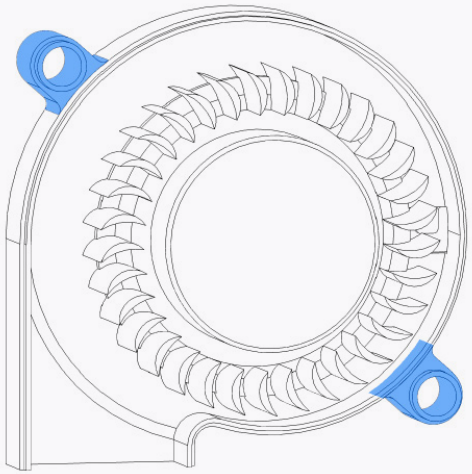
Mind the fan orientation.

FAN WIRE ROUTING

Route the fan wires as shown, exiting on the right side





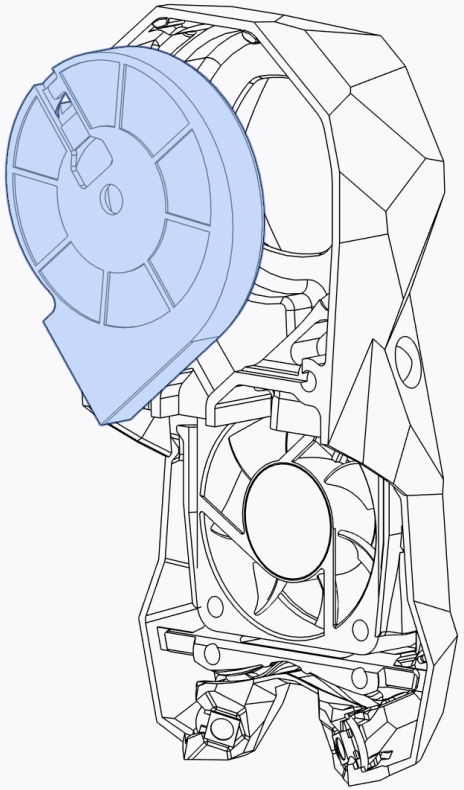


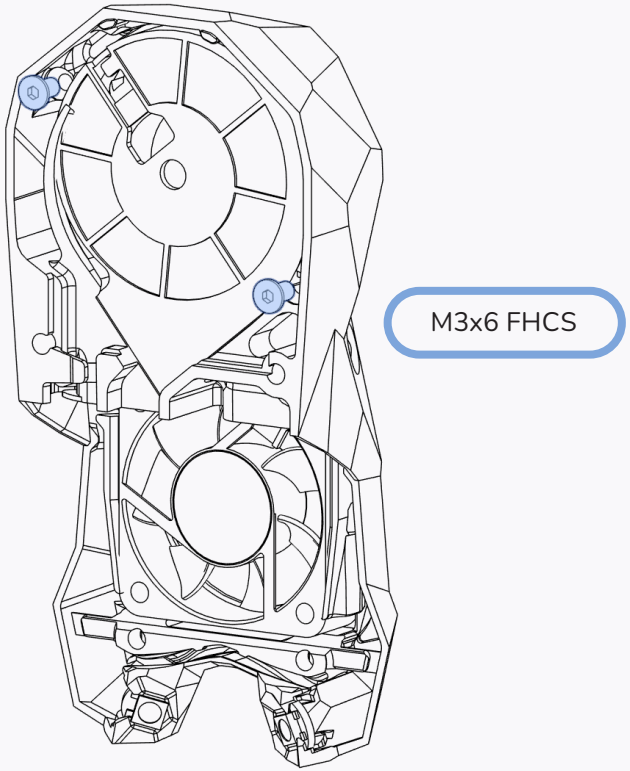
FAN PREPARATION

Remove the front of the 5015 fan. Clip off and file down the stock mounting ears.

There's a trimming jig STL included in the release to make this task easier.

You dont need to screw the 5015 fan into the SB Face, you will install the SB0000 Fan PCB there later.

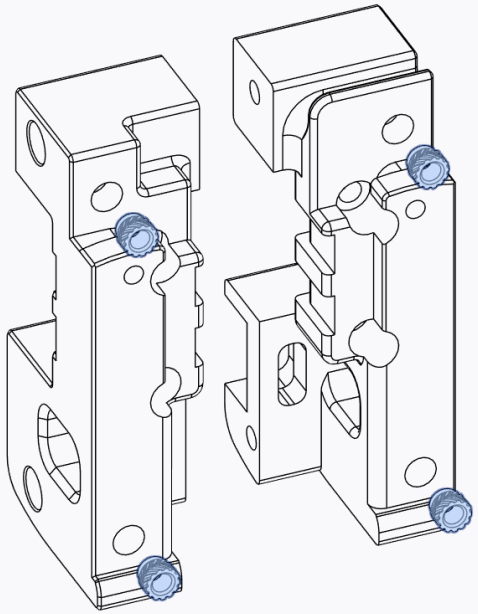




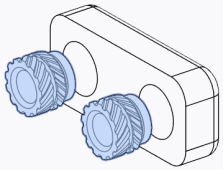
Yeah, it went about as expected.

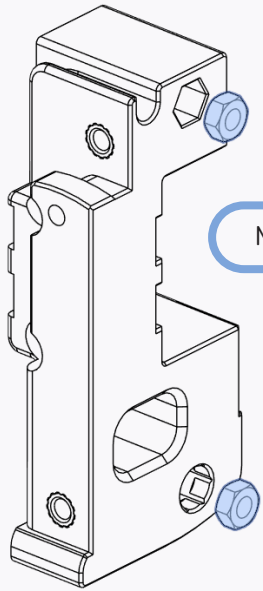
All parts of the Stealthburner are now pretty much finished. Now you will need to install the Can-Board on there and do some basic wiring. Switch to the SB2209 RP2040 Manual.





Heat Set Insert





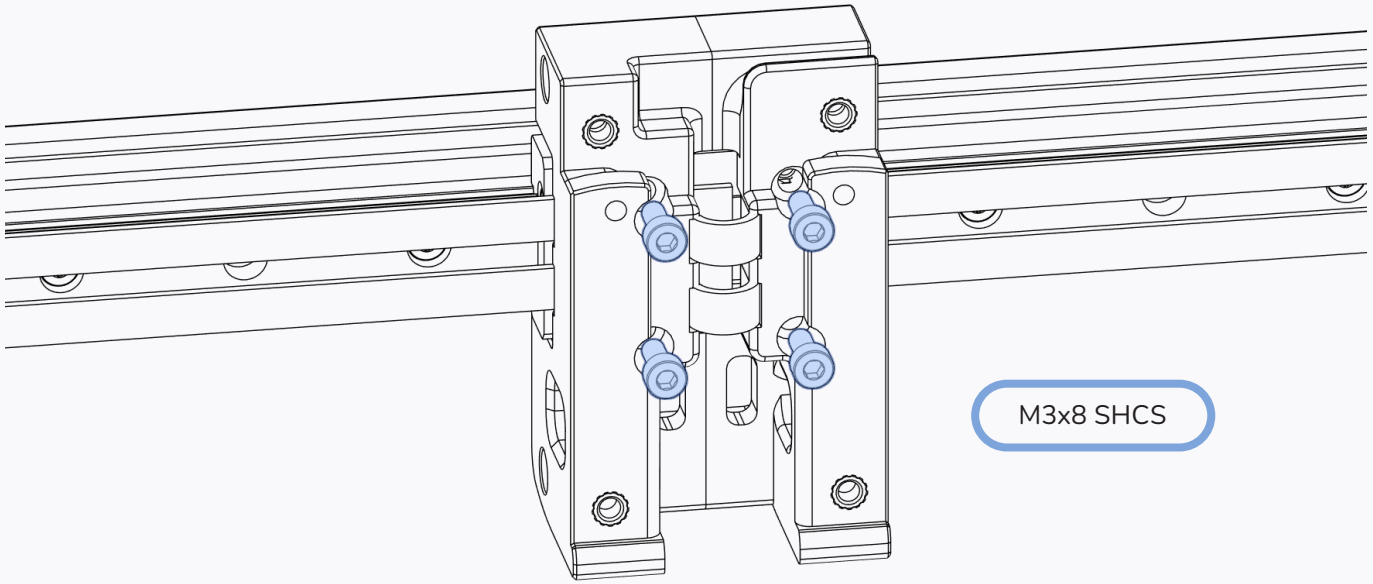
M3 Hex Nut

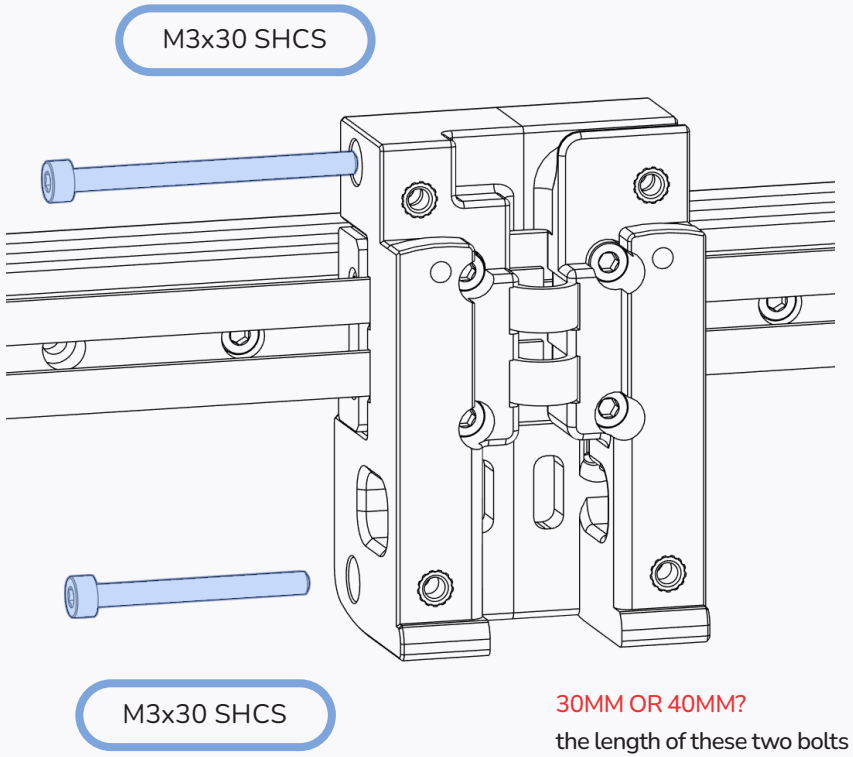
PRINTER SPECIFIC PARTS

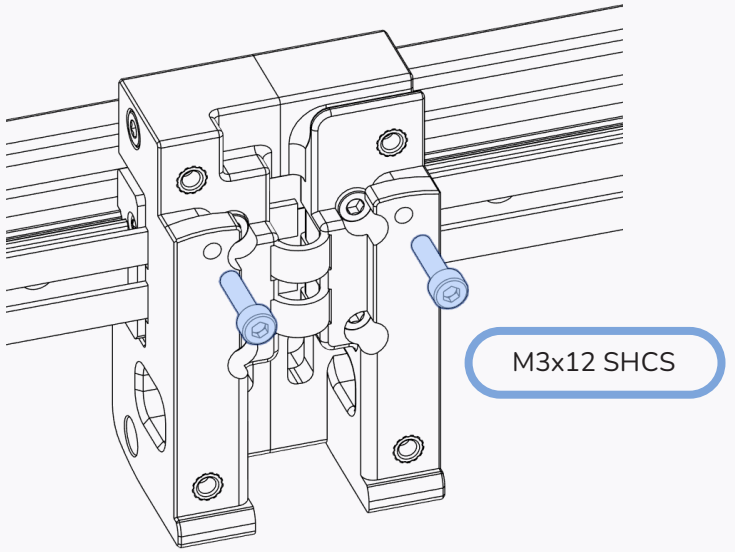
We provide different versions of the carriage to fit our different printer designs. Be sure to pick the right one for your printer. the next few pages may look slightly different if you are building a Switchwire printer.

FOLLOW BELTING AND X CARRIAGE INSTALLATION OUTLINED IN PRINTER MANUAL

Consult the printer's manual for instructions on how to run the belts and details on carriage mounting.

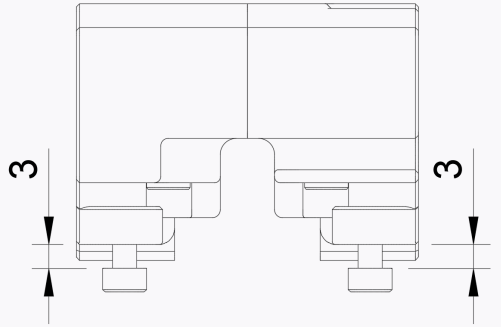


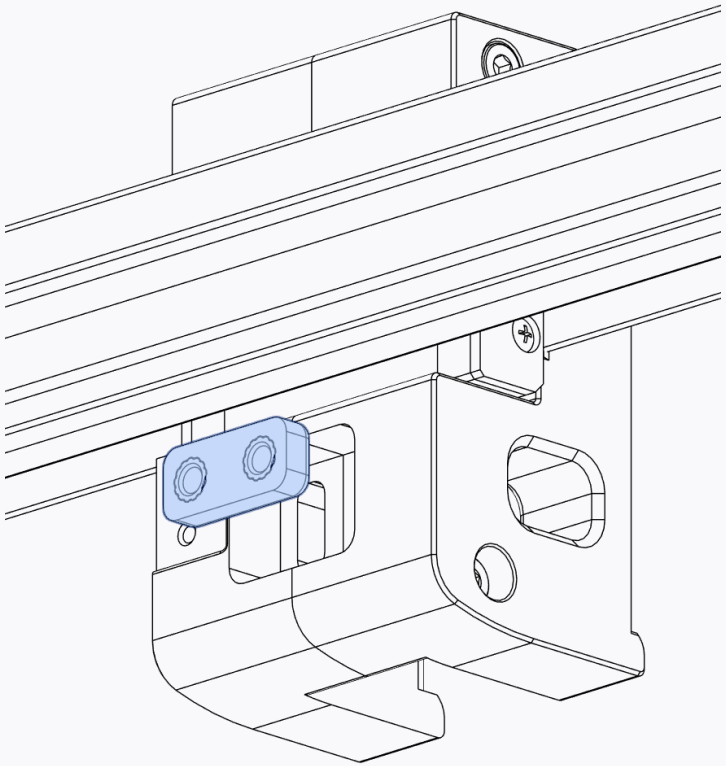


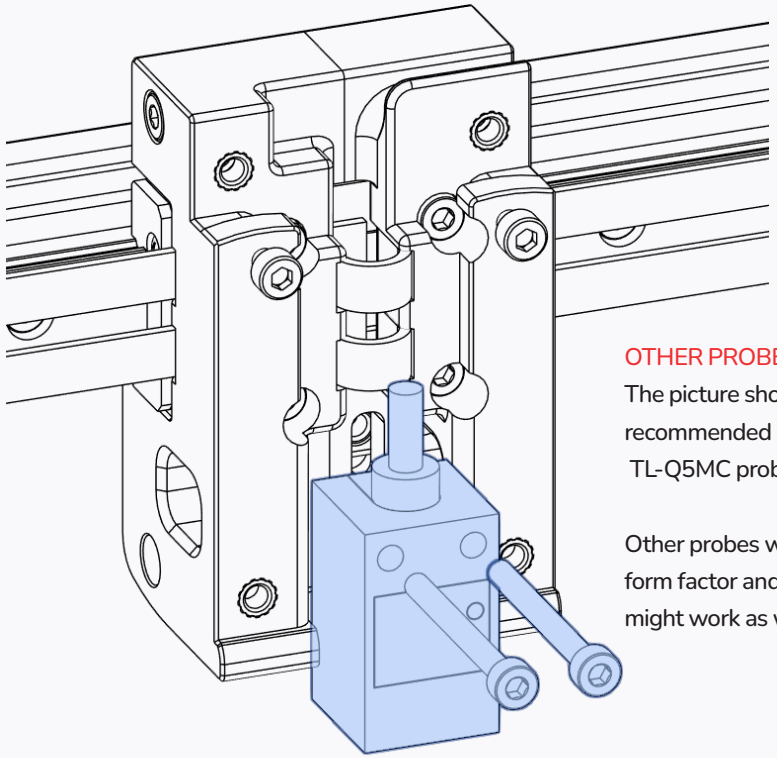


30MM OR 40MM?

the length of these two bolts can be different based on the specific printer you are building. This step has some overlap with the printer manuals. please refer to the printer psecific manual for the correct bolt lengths







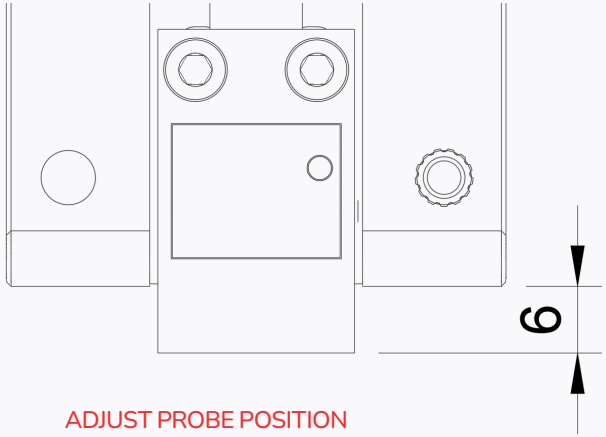
OTHER PROBE TYPES

The picture shows the recommended Omron TL-Q5MC probe.

Other probes with a similar form factor and characteristics might work as well.

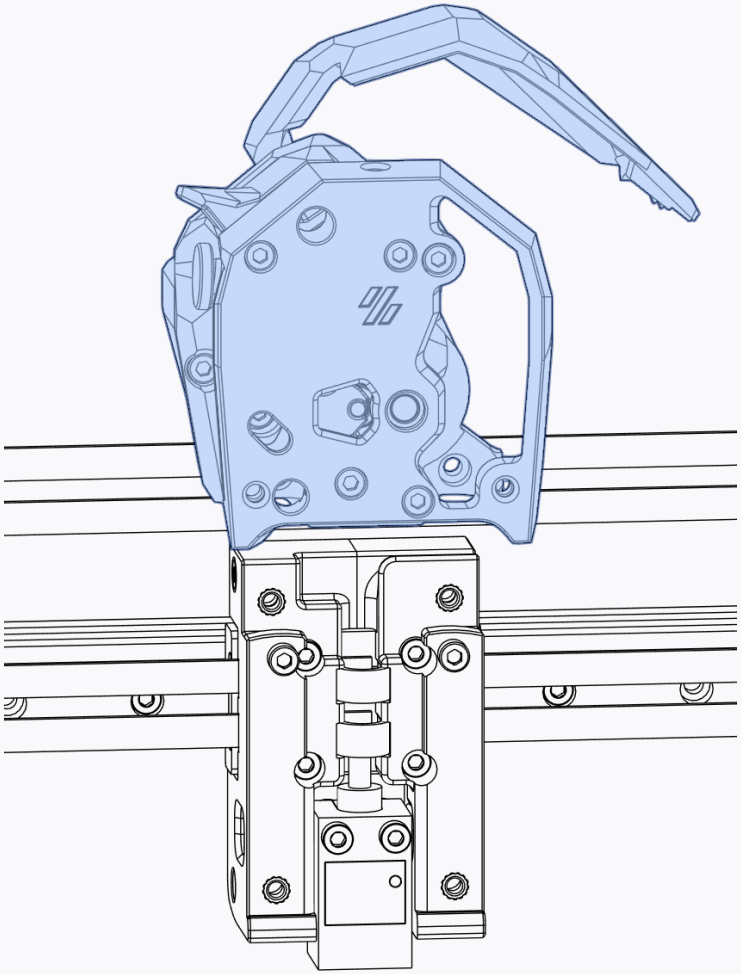
Omron Style Probe

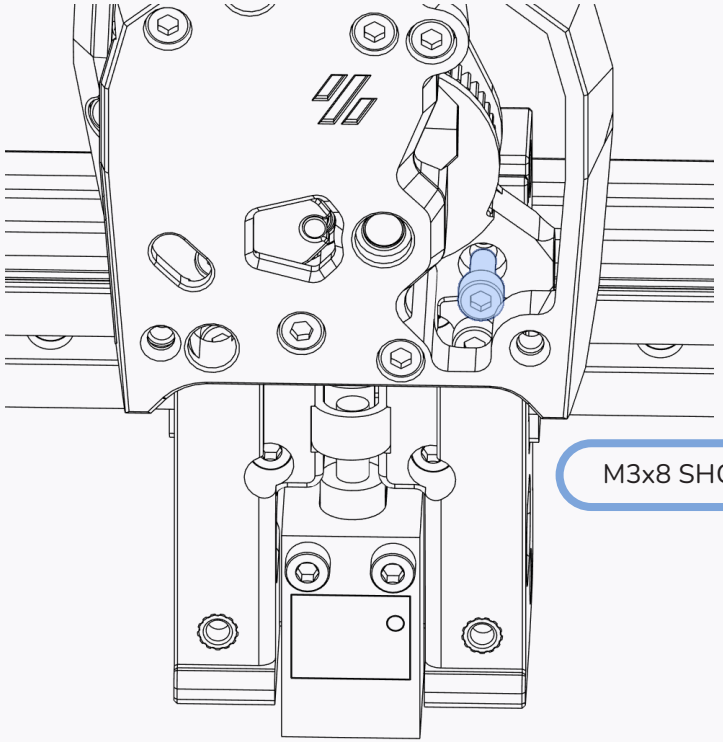
M3x30 SHCS

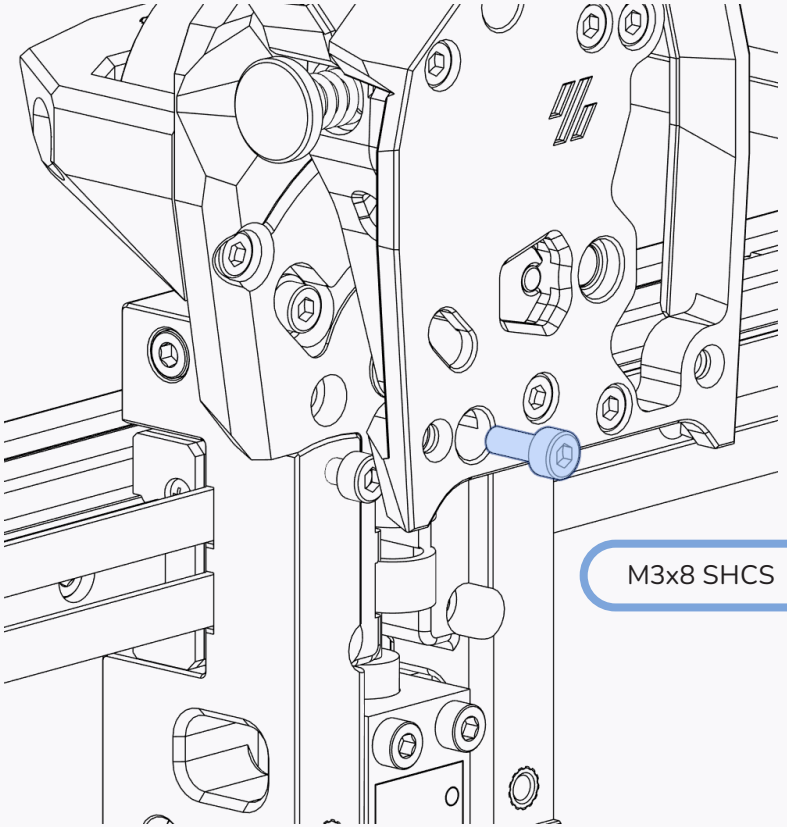


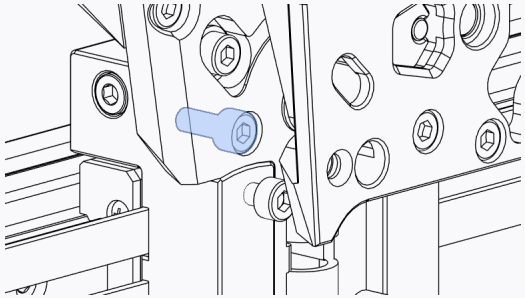
ADJUST PROBE POSITION

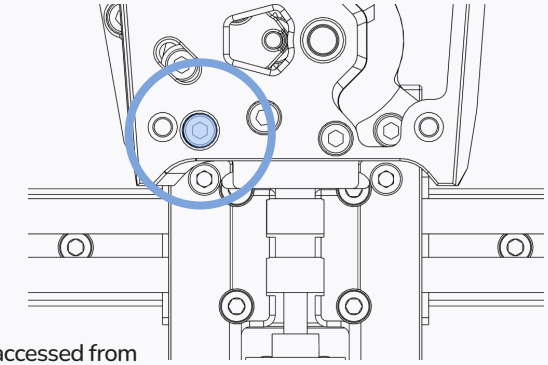
The position can be fine-tuned later.
Set an initial position of about 6mm
below the plastic part.




M3x8 SHCS

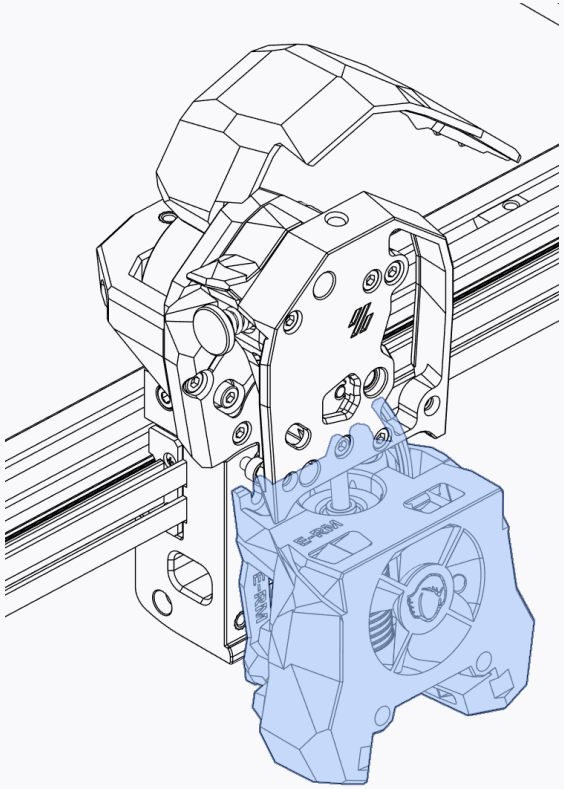


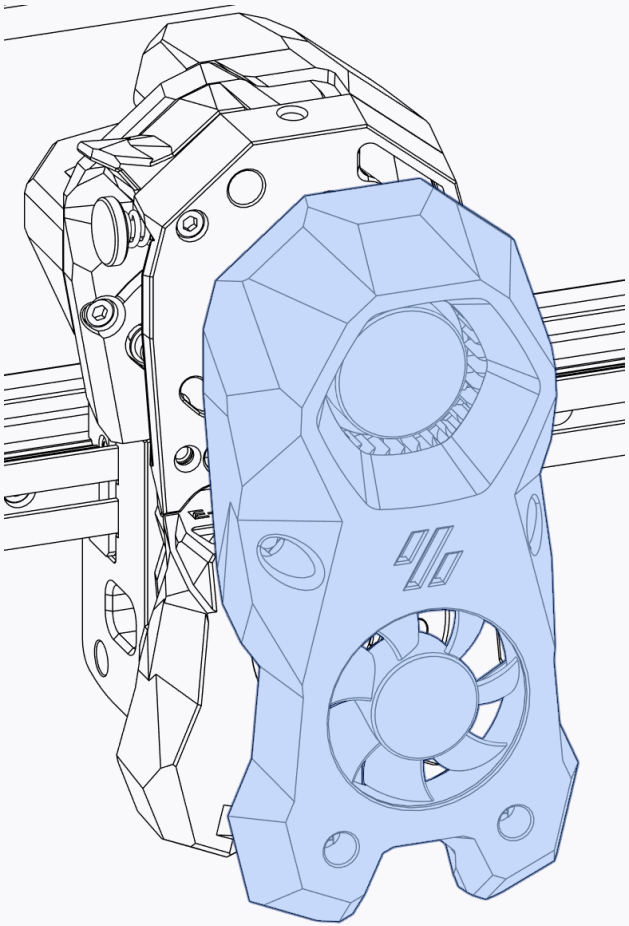


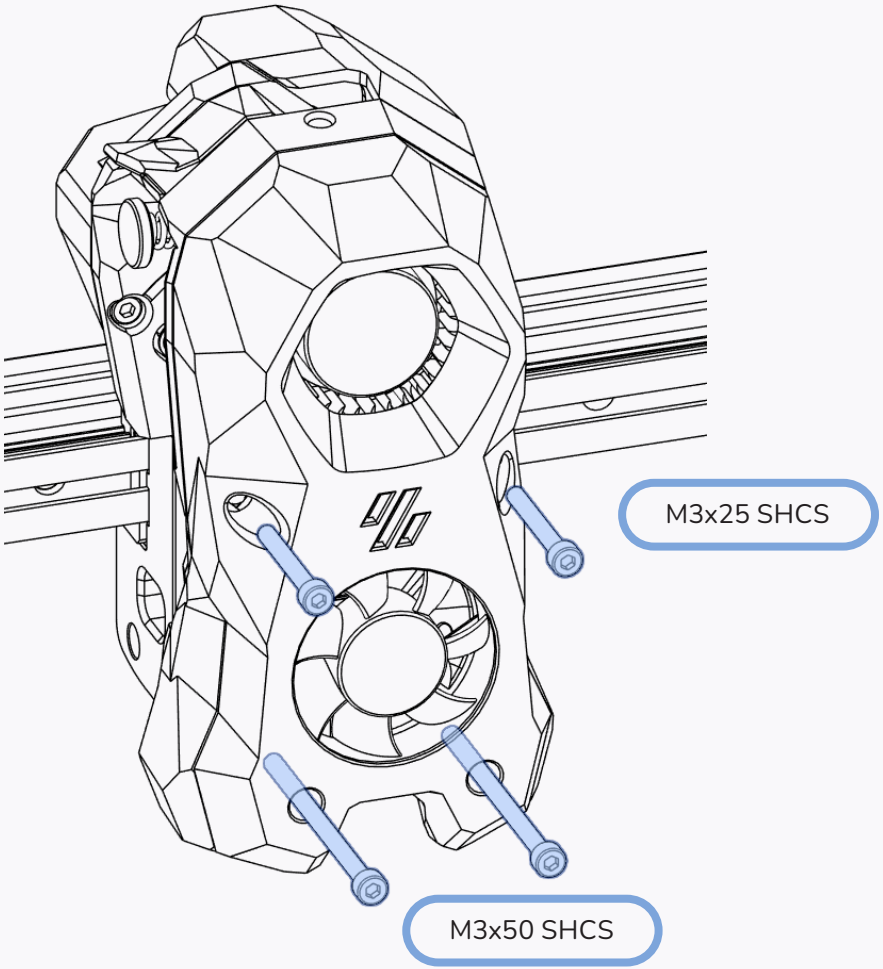


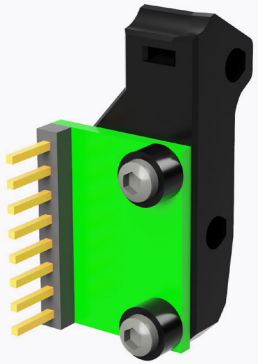
ACCESS HOLE

The bolt can be accessed from the front of the extruder.



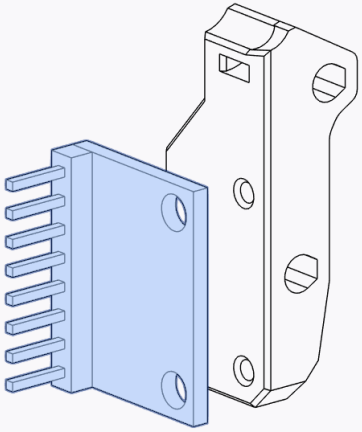



M3x25 SHCS
M3x50 SHCS

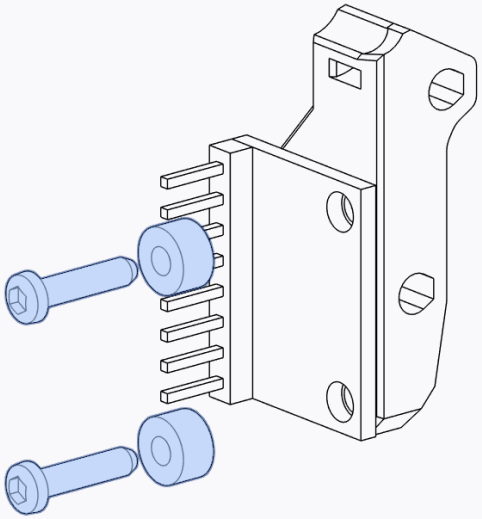


ADXL MOUNT AND INPUT SHAPER CALIBRATION

The ADXL is intended to be mounted only when doing vibration testing/input shaper calibration and should not be left on the tool head during normal printing.



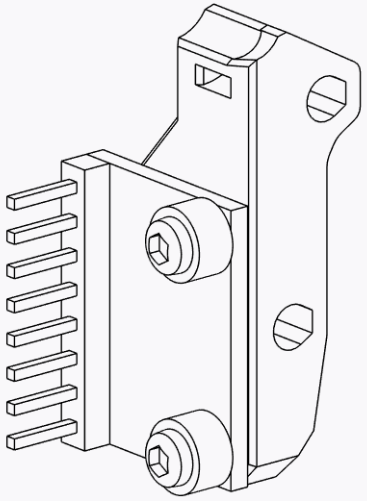
ADXL345 Prototype Board



M2x10 Self Tapping Screws

ISOLATION SPACERS

Most ADXL345 prototype boards have circuit tracks close to the mounting holes. Make sure to install spacers to prevent damage to the board. We included printable spacers in the release.



MOUNTING TO THE TOOLHEAD

You can use M3x12 screws to mount this assembly to your tooolhead.

ASSEMBLY COMPLETED! ... NEXT STEP: SETUP & CALIBRATION

This manual is designed to be a reference manual for the build process of a Voron StealthBurner toolhead. Additional details about the build and background on advanced topics can be found on our documentation page linked below.

The software setup and other initial setup steps with your new printer can also be found on our documentation page. We recommend starting here.



https://docs.vorondesign.com



https://github.com/VoronDesign/Voron-Stealthburner

HOW TO GET HELP

If you need assistance with your build, we're here to help. Head on over to our Discord group and post your questions. This is our primary medium to help VORON Users and we have a great community that can help you out if you get stuck. Alternatively, you can use our subreddit.



https://discord.gg/voron



https://www.reddit.com/r/VORONDesign

REPORTING ISSUES

Should you find an issue in this document or have a suggestion for an improvement please consider opening an issue on GitHub (https://github.com/VoronDesign/Voron-Stealthburner/issues).

When raising an issue please include the relevant page numbers and a brief description; annotated screenshots are also very welcome.

We periodically update the manual based on the feedback we get.



Website
vorondesign.com

Github
github.com/vorondesign

Docs
docs.vorondesign.com

Discord
discord.gg/voron

